

K.E.T's V.G.Vaze College of Arts, Science & Commerce

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Practical
1

Configure Routers

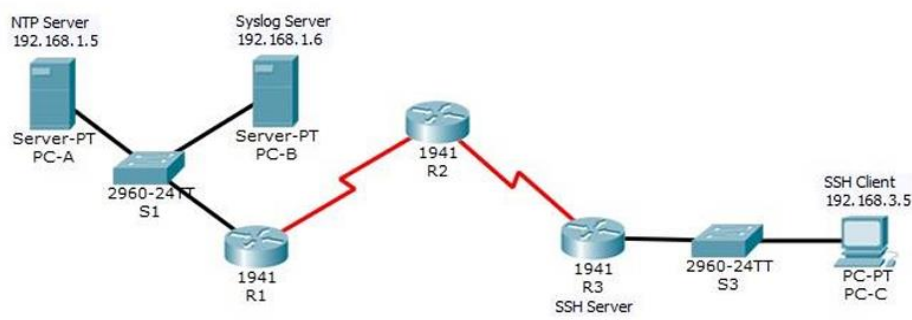
- a) OSPF MD5 authentication.
- b) NTP.
- c) To log messages to the syslog server.
- d) to support SSH connections.

8-12-25

8-12-25

15-12-25

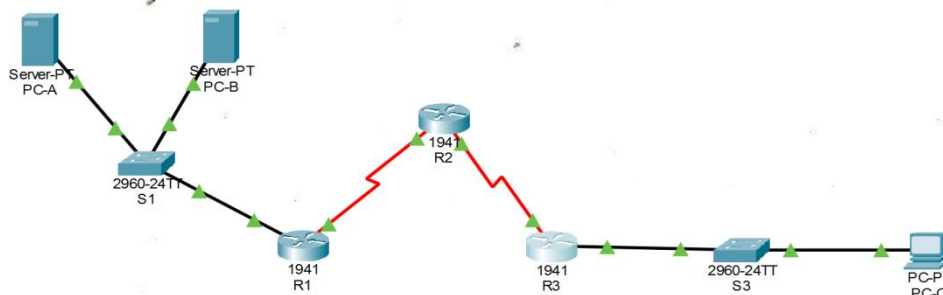
15-12-25



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.1	255.255.255.252	N/A	N/A
PC-A	NIC	192.168.1.5	255.255.255.0	192.168.1.1	S1 F0/6
PC-B	NIC	192.168.1.6	255.255.255.0	192.168.1.1	S2 F0/18
PC-C	NIC	192.168.3.5	255.255.255.0	192.168.3.1	S3 F0/18

Now the connectivity has been done Successfully.



Go to router 1> CLI>{enter}> and type the following command

```

R1>
R1>EN
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#network 192.168.1.0 0.0.0.255 area 0
R1(config-router)#network 10.1.1.0 0.0.0.3 area 0
R1(config-router)#
same go to router 2>CLI>
%Error parsing filename (Bad file number)
R2(config-if)#EX
R2(config)#EX
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#interface Serial0/0/0
R2(config-if)#router ospf 2
R2(config-router)#network 10.1.1.0 0.0.0.3 area 0
R2(config-router)#network
00:44:23: %OSPF-5-ADJCHG: Process 2, Nbr 192.168.1.1 on Serial0/0/0 from LOADING to FULL, Loading Done
Done

% Incomplete command.
R2(config-router)#network 10.2.2.0 0.0.0.3 area 0
R2(config-router)#

```

same way go to router 3>CLI>

```

R3>enable
R3#
R3#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#interface GigabitEthernet0/1
R3(config-if)#
?Bad filename
%Error parsing filename (Bad file number)
R3(config-if)#router ospf 3
R3(config-router)#network 192.168.3.0 0.0.0.255 area 0
R3(config-router)#network 10.2.2.0 0.0.0.3 area 0
R3(config-router)#
00:55:36: %OSPF-5-ADJCHG: Process 3, Nbr 10.2.2.2 on Serial0/0/0 from LOADING to FULL, Loading Done

```

Now the connection that was failed previously from PC-A to router 2 or is now Successful.

Realtime										
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC-A	R2	ICMP		0.000	N	7	(edit)	(delete)
	Successful	PC-A	R2	ICMP		0.000	N	8	(edit)	(delete)
	Successful	PC-A	R3	ICMP		0.000	N	9	(edit)	(delete)
	Successful	PC-A	PC-C	ICMP		0.000	N	10	(edit)	(delete)

Whether the connection is successful or not we can also check this through the cmd of PC-A.

PC-A

Physical Config Services Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.3.5

Pinging 192.168.3.5 with 32 bytes of data:

Reply from 192.168.3.5: bytes=32 time=2ms TTL=125
Reply from 192.168.3.5: bytes=32 time=2ms TTL=125
Reply from 192.168.3.5: bytes=32 time=29ms TTL=125
Reply from 192.168.3.5: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.3.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 29ms, Average = 8ms

C:\>
```

Same way we can check from PC-B>

PC-B

Physical Config Services Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.1.6

Pinging 192.168.1.6 with 32 bytes of data:

Reply from 192.168.1.6: bytes=32 time=9ms TTL=128
Reply from 192.168.1.6: bytes=32 time=14ms TTL=128
Reply from 192.168.1.6: bytes=32 time=6ms TTL=128
Reply from 192.168.1.6: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 14ms, Average = 7ms

C:\>|
```

a) OSPF MD5 authentication.

go to router 1> CLI >

```
% Invalid input detected at '^' marker.

R1>EN
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#area 0 authentication message-digest
R1(config-router)#
```

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go to router 2>

```

00:58:24: %OSPF-5-ADJCHG: Process 2, Nbr 192.168.1.1 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Dead timer expired

00:58:24: %OSPF-5-ADJCHG: Process 2, Nbr 192.168.1.1 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Interface down or detached

% Incomplete command.
R2(config)#router ospf 2
R2(config-router)#area 0 authentication message-digest
R2(config-router)#

```

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same in router 3>

```

R3(config)#router
01:09:26: %OSPF-5-ADJCHG: Process 3, Nbr 10.2.2.2 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Dead timer expired

01:09:26: %OSPF-5-ADJCHG: Process 3, Nbr 10.2.2.2 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Interface down or detached

% Incomplete command.
R3(config)#router ospf 3
R3(config-router)#area 0 authentication message-digest
R3(config-router)#
01:10:06: %OSPF-5-ADJCHG: Process 3, Nbr 10.2.2.2 on Serial0/0/0 from LOADING to FULL, Loading Done

```

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Now again go to router 1>

R1

Physical Config CLI Attributes

IOS Command Line Interface

```

R1>
R1>conf t
^
% Invalid input detected at '^' marker.

R1>config terminal
^
% Invalid input detected at '^' marker.

R1>EN
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#router ospf 1
R1(config-router)#area 0 authentication message-digest
R1(config-router)#
01:08:16: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Dead timer expired

01:08:16: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Interface down or detached

01:09:46: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/0 from LOADING to FULL, Loading Done

R1(config-router)#interface s0/0/0
R1(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
R1(config-if)#
R1#
%SYS-5-CONFIG_1: Configured from console by console

01:15:46: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Dead timer expired

01:15:46: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Interface down or detached

```

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Now in router 2>

```

01:05:45: %OSPF-5-ADJCHG: Process 2, Nbr 192.168.1.1 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Dead timer expired

01:05:45: %OSPF-5-ADJCHG: Process 2, Nbr 192.168.1.1 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Interface down or detached

R2(config-router)#interface s0/0/0
R2(config-if)#ip ospf message-digest-key 1 md5 MD5pa55
R2(config-if)#
01:08:08: %OSPF-5-ADJCHG: Process 2, Nbr 192.168.1.1 on Serial0/0/0 from LOADING to FULL, Loading Done

R2(config-if)#interface s0/0/1
R2(config-if)#ip ospf message-digest-key 1 md5 MDpa55
R2(config-if)#

```

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Top

Now in router 3>

```

R3(config-router)#interface s0/0/1
R3(config-if)#ip ospf message-digest-key 1 md5 MDpa55
R3(config-if)#

```

The final step is to verify> go to router 1 >CLI>

```
R1#show ip ospf interface
```

In same way we can check for Router 2 and router 3.

B)NTP

Go in PC-A > services >

PC-A Services configuration window. The NTP service is highlighted in the left sidebar and is set to 'On' in the main panel. The authentication section shows 'Enable' selected, 'Key' set to '1', and 'Password' set to 'NTPpa55'. Below this is a calendar for February 2026, with the 7th of the month highlighted in blue.

Now after this go to router 1 > CLI

```

R1>EN
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ntp server 192.168.1.6
R1(config)#nt
% Unknown command or computer name, or unable to find computer address

R1(config)#ntp update-calender
% Invalid input detected at '^' marker.

R1(config)#
Next > go to router 2>

```

```

R2>EN
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ntp server 192.168.1.5
R2(config)#ntp update-calender
      ^
% Invalid input detected at '^' marker.

R2(config)#ntp update-calendar
R2(config)#

```

Same for Router 3>

```

R3>EN
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ntp server 192.168.1.5
R3(config)#ntp update-calendar
R3(config)#

```

Again go to router 1 >CLI

```

R1(config)#ntp authenticate
R1(config)#ntp trusted-key 1
R1(config)#ntp authentication-key 1 md5 NTPpa55
      ^
% Invalid input detected at '^' marker.

R1(config)#ntp authentication-key 1 md5 NTPpa55
R1(config)#service timestamps log datetime msec
R1(config)#

```

Again we have to do same for Router 2 and 3.

```

R2(config)#ntp authenticate
R2(config)#ntp trusted-key 1
R2(config)#ntp authentication-key 1 md5 NTPpa55
R2(config)#service timestamps log datetime msec
R2(config)#

```

```

R3(config)#ntp authenticate
R3(config)#ntp trusted-key 1
R3(config)#ntp authentication-key 1 md5 NTPpa55
R3(config)#service timestamps log datetime msec
R3(config)#

```

Click on 

Now to see the output of previous three image then come to Router# > by Exit and then type (show clock) now you will be able to see the current time.

C)To log messages to the syslog server.
Before: in PC-B > services > SYSLOG >

Nothing will be there.

Go to router 1 > CLI>

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#logging host 192.168.1.6
R1(config)#
```

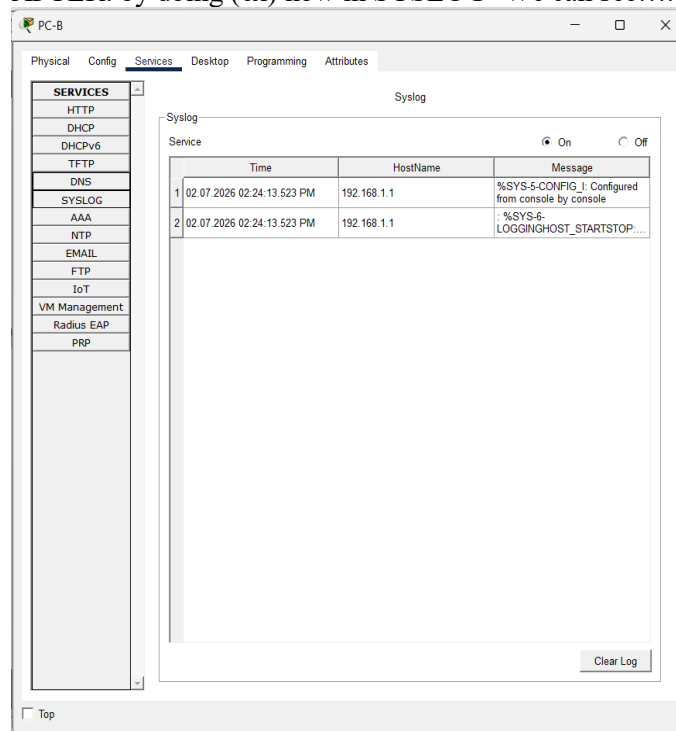
Do this same for router 2 and 3.

```
R2>EN
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#logging host 192.168.1.6
R2(config)#
```

```
R3>EN
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#logging host 192.168.1.6
R3(config)#
```

To see the Result type (ex) > here and same for router 2 and 3

AFTER: by doing (ex) now in SYSLOG>We can see....



Now by going in router 1 ,2and 3 type command >show logging > will show the logging info >

```

- CLI initiated
R1#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
                  0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 17 messages logged, xml disabled,
                  filtering disabled
Monitor logging: level debugging, 17 messages logged, xml disabled,
                  filtering disabled
Buffer logging: level debugging, 0 messages logged, xml disabled,
                  filtering disabled

Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

ESM: 0 messages dropped
Trap logging: level informational, 17 message lines logged

```

D)to support SSH connections.

Go in router 3 > CLI>

```

R3(config)#ip domain-name ccnasecurity.com
R3(config)#username SSHadmin privilege 15 secret ciscosshpa55

% Invalid input detected at '^' marker.

R3(config)#username SSHadmin privilege 15 secret ciscosshpa55
R3(config)#line vty 0 4
R3(config-line)#login local
R3(config-line)#transport input ssh
R3(config-line)#ex
% Ambiguous command: "ex"
R3(config-line)#en
% Ambiguous command: "en"
R3(config)#crypto key zeroize rsa
% No Signature RSA Keys found in configuration.

R3(config)#crypto key generate rsa
The name for the keys will be: R3.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R3(config)#ex
*Mar 1 2:40:12.427: %SSH-5-ENABLED: SSH 1.99 has been enabled
R3#
*Mar 01, 02:40:30.4040: SYS-5-CONFIG_I: Configured from console by console
R3#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip ssh time-out 90
R3(config)#ip ssh authentication-retries 2
R3(config)#ip ssh version 2
R3(config)#

```

After this go in PC-C>desktop>Cmd> (password:ciscosshpa55)

```

Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 192.168.3.1
Trying 192.168.3.1 ...Open

[Connection to 192.168.3.1 closed by foreign host]
C:\>ssh -l SSHAdmin 192.168.3.1
Invalid Command.

C:\>ssh -l SSHAdmin 192.168.3.1

Password:
% Password: timeout expired!
% Login invalid

[Connection to 192.168.3.1 closed by foreign host]
C:\>ssh -l SSHAdmin 192.168.3.1

Password:

R3#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 90 secs; Authentication retries: 2
R3#

```

Then go in router 2> CLI

```

R2#ssh -v 2 -l SSHAdmin 10.2.2.1

Password:

R3#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 90 secs; Authentication retries: 2
R3#exit

[Connection to 10.2.2.1 closed by foreign host]
R2#

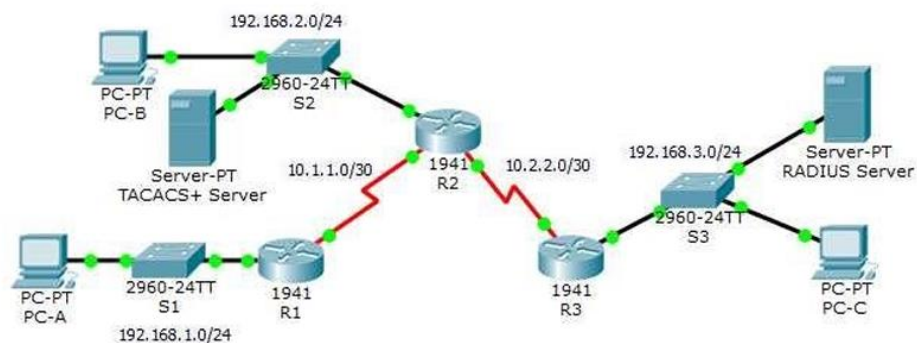
```

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2

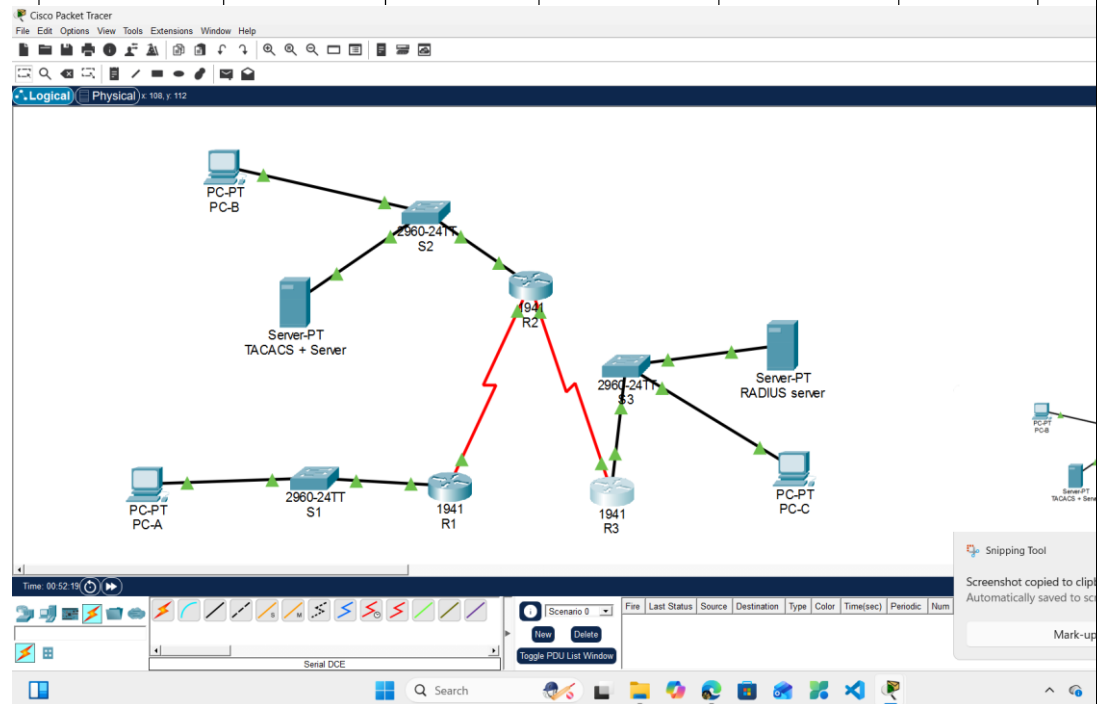
Configure AAA Authentication

5-01-26

- Configure a local user account on Router and configure authenticate on the console and vty lines using local AAA
- Verify local AAA authentication from the Router console and the PC-A client



Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/1
	S0/0/0 (DCE)	10.1.1.2	255.255.255.252	N/A	N/A
R2	G0/0	192.168.2.1	255.255.255.0	N/A	S2 F0/2
	S0/0/0	10.1.1.1	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.1	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.2	255.255.255.252	N/A	N/A
TACACS+ Server	NIC	192.168.2.2	255.255.255.0	192.168.2.1	S2 F0/6
RADIUS Server	NIC	192.168.3.2	255.255.255.0	192.168.3.1	S3 F0/1
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/2
PC-B	NIC	192.168.2.3	255.255.255.0	192.168.2.1	S2 F0/1
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18



Now in router 1>CLI>

```

%Error parsing filename (Bad file number)
R1(config-if)#EN
% Ambiguous command: "EN"
R1(config)#line vty 0 4
R1(config-line)#password admin
R1(config-line)#login
R1(config-line)#en
% Ambiguous command: "en"
R1(config)#ospf 1
^
% Invalid input detected at '^' marker.

R1(config)#network ospf 1
^
% Invalid input detected at '^' marker.

R1(config)#router ospf 1
R1(config-router)#network 192.168.1.0 0.0.0.255 area 0
R1(config-router)#network 10.1.1.0 0.0.0.255 area 0
R1(config-router)#

```

Same for router 2 and 3

```

R2(config-if)#en
% Ambiguous command: "en"
R2(config)#line vty 0 4
R2(config-line)#password admin
R2(config-line)#login
R2(config-line)#en
% Ambiguous command: "en"
R2(config)#router ospf 2
R2(config-router)#network 192.168.2.0 0.0.0.255 area 0
R2(config-router)#network 10.2.2.0 0.0.0.255 area 0
R2(config-router)#network 10.1.1.0 0.0.0.255 area 0
R2(config-router)#
00:31:56: %OSPF-5-ADJCHG: Process 2, Nbr 192.168.1.1 on Serial0/0/0 from LOADING to FULL, Load
Done

```

Copy

```

R3(config-if)#en
% Ambiguous command: "en"
R3(config)#line vty 0 4
R3(config-line)#password admin
R3(config-line)#login
R3(config-line)#en
% Ambiguous command: "en"
R3(config)#router ospf 3
R3(config-router)#network 192.168.3.0 0.0.0.255 area 0
R3(config-router)#network 10.2.2.0 0.0.0.255 area 0
R3(config-router)#

```

Configuring AAA:
Go in router 1>CLI

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#username Admin1 secret admin1pa55
R1(config)#aaa new-model
R1(config)#aaa authentication login default local
R1(config)#line console 0
R1(config-line)#login authentication default
R1(config-line)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
ex
```

User Access Verification

Username: Admin1

In above username : Admin1 ,password:admin1pa55

After this again come in router 1 >

```
R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip domain-name ccnasecurity.com
R1(config)#crypto key generate rsa
The name for the keys will be: R1.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#aaa authentication login SSH-LOGIN local
*Mar 1 0:44:28.115: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#line vty 0 4
R1(config-line)#login authentication SSH-LOGIN
R1(config-line)#transport input ssh
R1(config-line)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
exit
```

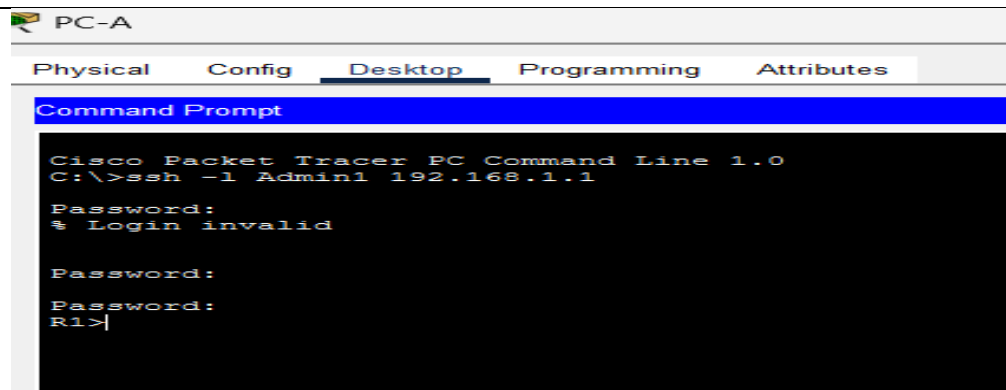
User Access Verification

Username: Admin1

Password:

R1>

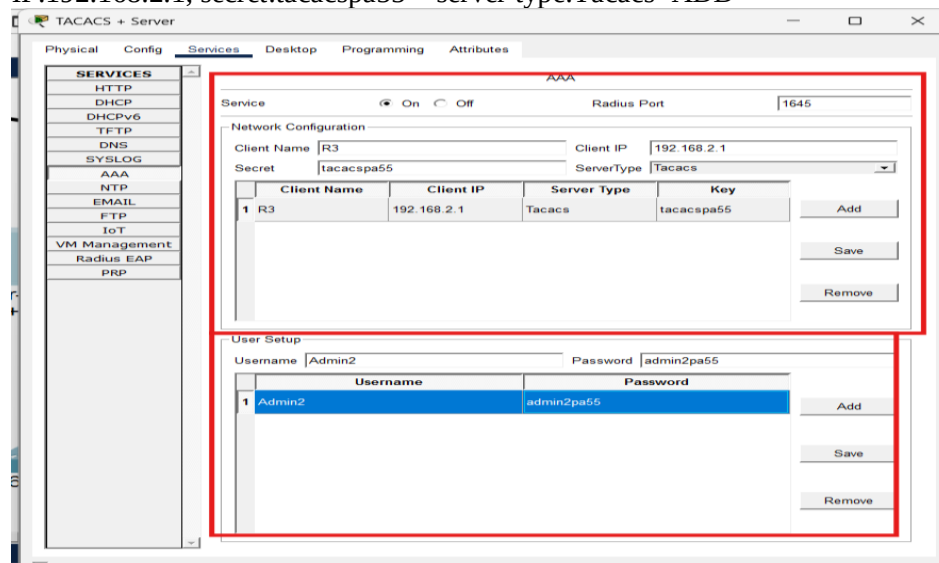
After this come in PC-A >cmd>password:admin1pa55



Now go in router 2 > CLI

```
R2>en
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#username Admin2 secret admin2pa55
R2(config)#
```

After this go in TACACS server > services > AAA > ON > client name:R2, IP:192.168.2.1, secret:tacacspa55 > server type:Tacacs>ADD



Agin go in router 2>CLI>

```
R2>en
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#username Admin2 secret admin2pa55
R2(config)#aaa new-model
R2(config)#aaa authentication login default group tacacs local
R2(config)#line console 0
R2(config-line)#login authentication default
R2(config-line)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console
exit
```

The down the username:Admin2, password:admin2pa55

User Access Verification

```
Username: Admin2
Password:
R2>
```

Successfully Done.

Now for router 3 > Cli>

```
R3>en
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#username Admin3 secret admin3pa55
R3(config)#
```

Now go in radius server > services > AAA>

The screenshot shows the 'RADIUS server' configuration window with the 'Services' tab selected. The 'AAA' service is configured with 'On' status and 'Radius Port' set to 1645. The 'Network Configuration' section shows a table with one entry: 'R3' with Client IP '192.168.3.1' and Key 'radiuspa55'. The 'User Setup' section shows a table with one entry: 'Admin3' with Password 'admin3pa55'.

Client Name	Client IP	Server Type	Key
1 R3	192.168.3.1	Radius	radiuspa55

Username	Password
1 Admin3	admin3pa55

Row again go in router 3 > cLI>continue>

```

R3(config)#aaa new-model
R3(config)#aaa authentication login default group radius local
R3(config)#line console 0
R3(config-line)#login authentication default
R3(config-line)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console
exit

```

Down password is :admin3pa55

User Access Verification

```

Username: Admin3
Password:
R3>

```

SuccessFullyDone .

Practical 3

Objectives:

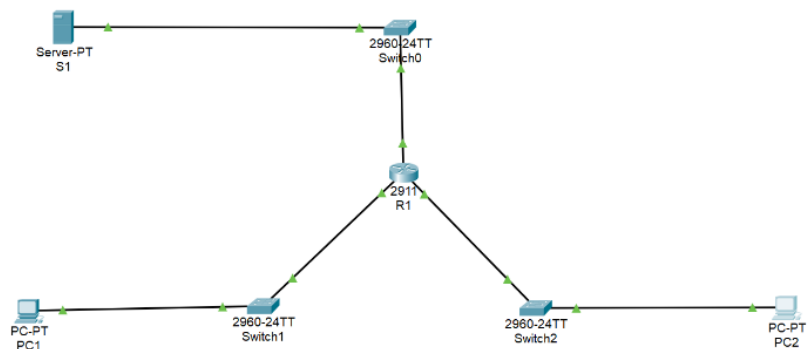
19-1-26

- Configure, Apply and Verify an Extended Numbered ACL
- Configure, Apply and Verify an Extended Named ACL

Scenario 1:

- PC1 should be allowed only FTP access
- PC2 should be allowed only web access
- Both PCs must ping server but not each others

Topology:



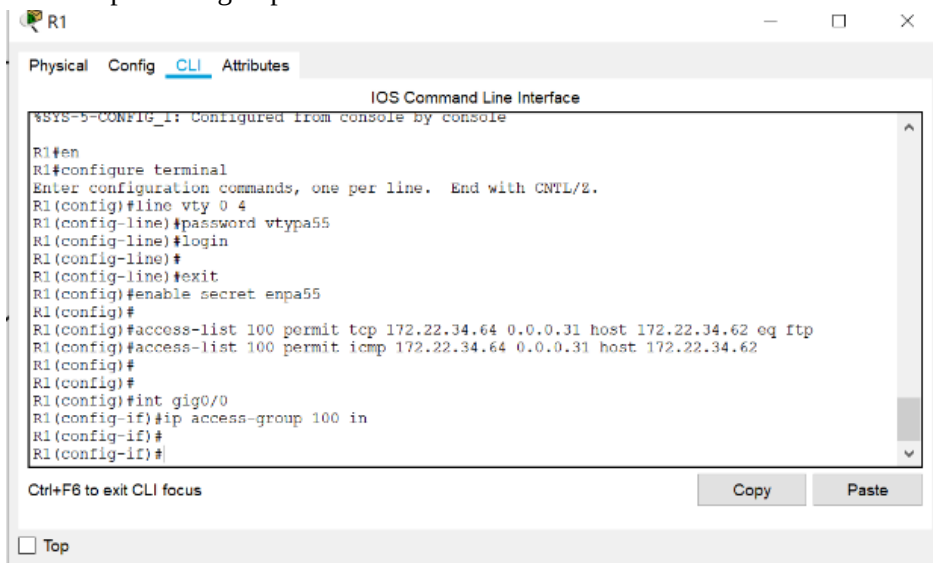
Connect all the devices as shown in the topology

Addressing table :

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	172.22.34.65	255.255.255.224	N/A
	G0/1	172.22.34.97	255.255.255.240	N/A
	G0/2	172.22.34.1	255.255.255.192	N/A
Server	NIC	172.22.34.62	255.255.255.192	172.22.34.1
PC1	NIC	172.22.34.66	255.255.255.224	172.22.34.65
PC2	NIC	172.22.34.98	255.255.255.240	172.22.34.97

Now Go to the CLI of R1 router and type the following commands

- Line vty 0 4
- Password vtyp55
- Login
- Exit
- Enable secret enpa55
- access-list 100 permit tcp 172.22.34.64 0.0.0.31 host 172.22.34.62 eq ftp
- access-list 100 permit icmp 172.22.34.64 0.0.0.31 host 172.22.34.62
- int gig0/0
- ip access-group 100 in



```

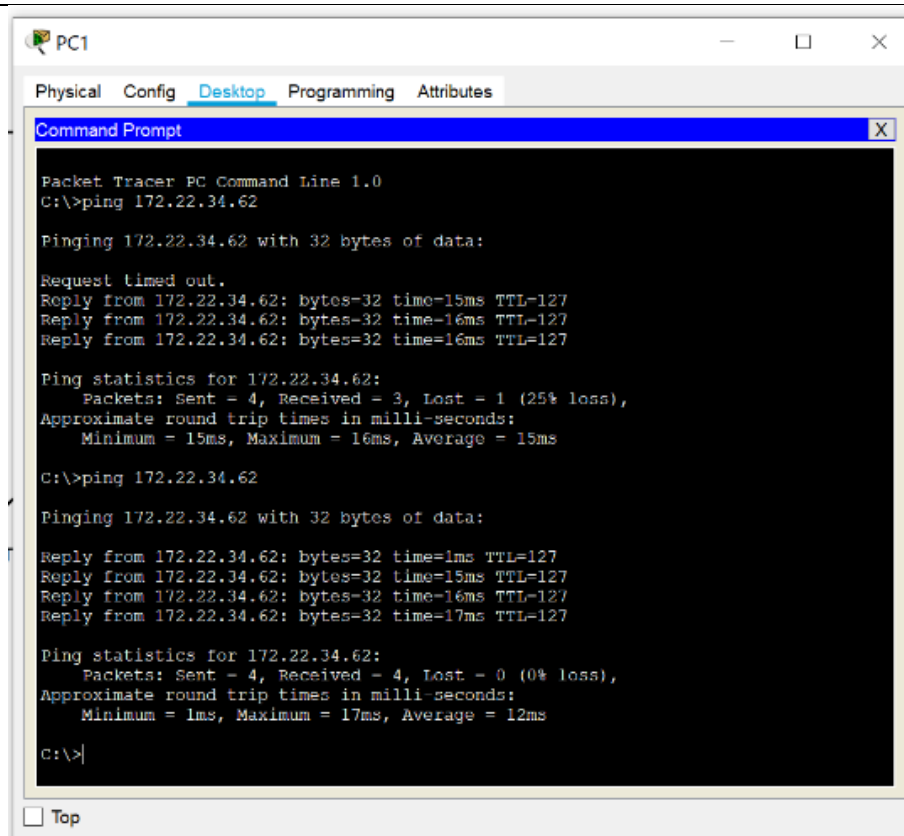
R1
R1#en
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#line vty 0 4
R1(config-line)#password vtyp55
R1(config-line)#login
R1(config-line)#
R1(config-line)#exit
R1(config)#enable secret enpa55
R1(config)#
R1(config)#access-list 100 permit tcp 172.22.34.64 0.0.0.31 host 172.22.34.62 eq ftp
R1(config)#access-list 100 permit icmp 172.22.34.64 0.0.0.31 host 172.22.34.62
R1(config)#
R1(config)#
R1(config)#int gig0/0
R1(config-if)#ip access-group 100 in
R1(config-if)#
R1(config-if)#

```

After saving the commands go to command prompt of PC1 and pass the command as:

- ping 172.22.34.62(Server)

First time it will not be successful but after that next time it will be successful



```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Packet Tracer PC Command Line 1.0
C:\>ping 172.22.34.62

Pinging 172.22.34.62 with 32 bytes of data:

Request timed out.
Reply from 172.22.34.62: bytes=32 time=15ms TTL=127
Reply from 172.22.34.62: bytes=32 time=16ms TTL=127
Reply from 172.22.34.62: bytes=32 time=16ms TTL=127

Ping statistics for 172.22.34.62:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 15ms, Maximum = 16ms, Average = 15ms

C:\>ping 172.22.34.62

Pinging 172.22.34.62 with 32 bytes of data:

Reply from 172.22.34.62: bytes=32 time=1ms TTL=127
Reply from 172.22.34.62: bytes=32 time=15ms TTL=127
Reply from 172.22.34.62: bytes=32 time=16ms TTL=127
Reply from 172.22.34.62: bytes=32 time=17ms TTL=127

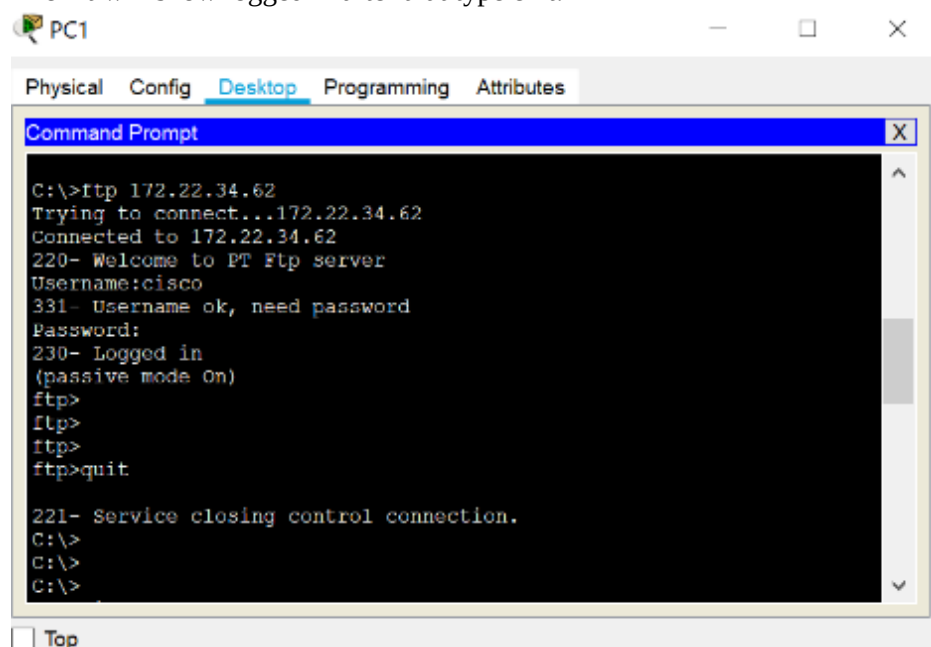
Ping statistics for 172.22.34.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 17ms, Average = 12ms

C:\>
```

Now pass the command as

- [ftp 172.22.34.62](#)
- Username as password will be “cisco”

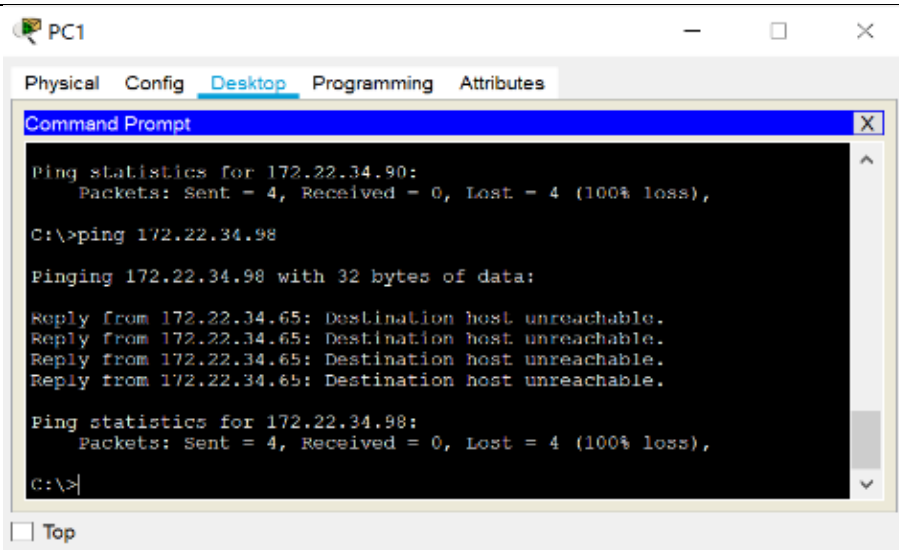
Then it will show logged in after that type exit.



```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ftp 172.22.34.62
Trying to connect...172.22.34.62
Connected to 172.22.34.62
220- Welcome to PT Ftp server
Username:cisco
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>
ftp>
ftp>quit

221- Service closing control connection.
C:\>
C:\>
C:\>
```

Now try to ping PC2 it should give output as destination host is unreachable(ping 172.22.34.98)



```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Ping statistics for 172.22.34.90:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 172.22.34.98

Pinging 172.22.34.98 with 32 bytes of data:

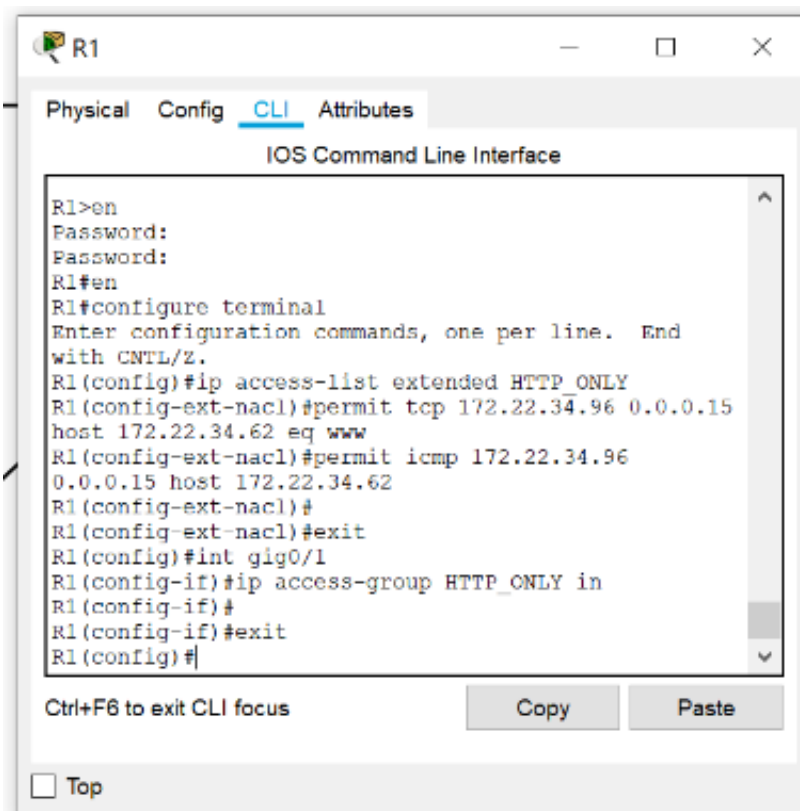
Reply from 172.22.34.65: Destination host unreachable.
Reply from 172.22.34.65: Destination host unreachable.
Reply from 172.22.34.65: Destination host unreachable.
Reply from 172.22.34.65: Destination host unreachable.

Ping statistics for 172.22.34.98:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Now go to R1 CLI and give commands as:

- En
- Password: enpa55
- Configure terminal
- Ip access-list extended HTTP_ONLY
- Permit tcp 172.22.34.96 0.0.0.15 host 172.22.34.62 eq www
- Permit icmp 172.22.34.96 0.0.0.15 host 172.22.34.62
- Exit
- Int gig0/1
- Ip access-group HTTP_ONLY in
- exit

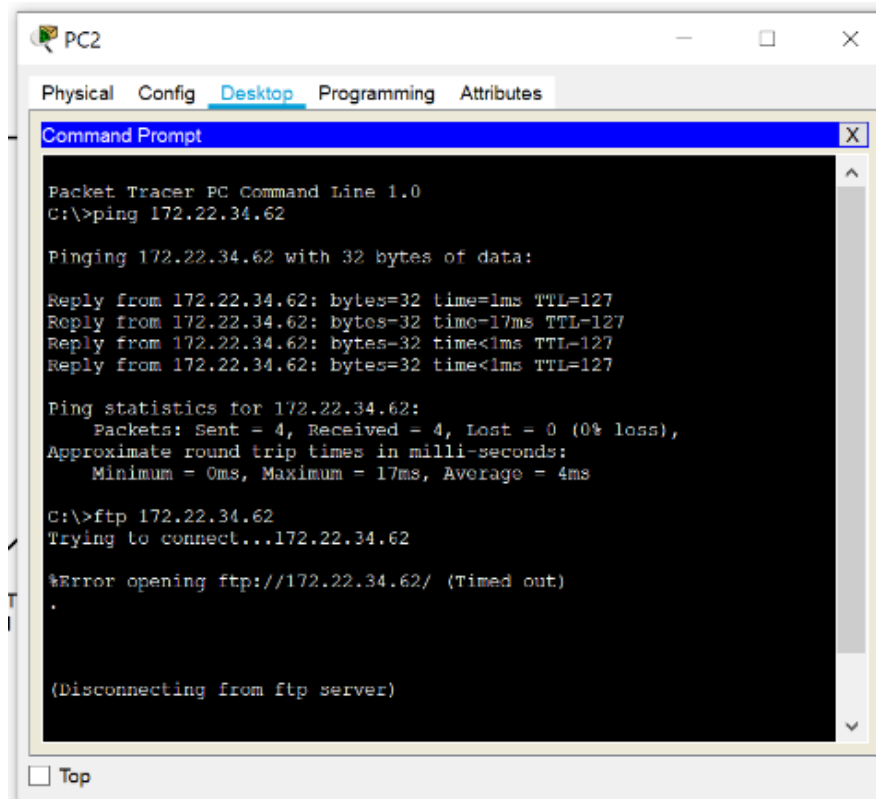


```
R1
Physical Config CLI Attributes
IOS Command Line Interface

R1>en
Password:
Password:
R1#en
R1#configure terminal
Enter configuration commands, one per line. End
with CNTL/Z.
R1(config)#ip access-list extended HTTP_ONLY
R1(config-ext-nacl)#permit tcp 172.22.34.96 0.0.0.15
host 172.22.34.62 eq www
R1(config-ext-nacl)#permit icmp 172.22.34.96
0.0.0.15 host 172.22.34.62
R1(config-ext-nacl)#
R1(config-ext-nacl)#exit
R1(config)#int gig0/1
R1(config-if)#ip access-group HTTP_ONLY in
R1(config-if)#
R1(config-if)#exit
R1(config)#
```

Now Go to the Command Prompt of PC2 and give the command as:

- ping 172.22.34.62(Server)
- when you give the command [ftp 172.22.34.62](ftp://172.22.34.62) it should give error



The screenshot shows a Packet Tracer PC window for PC2. The 'Desktop' tab is selected, and a 'Command Prompt' window is open. The command prompt displays the results of a ping command to 172.22.34.62, which was successful. It then shows an attempt to connect via ftp to 172.22.34.62, which failed with a 'Timed out' error.

```
Packet Tracer PC Command Line 1.0
C:\>ping 172.22.34.62

Pinging 172.22.34.62 with 32 bytes of data:

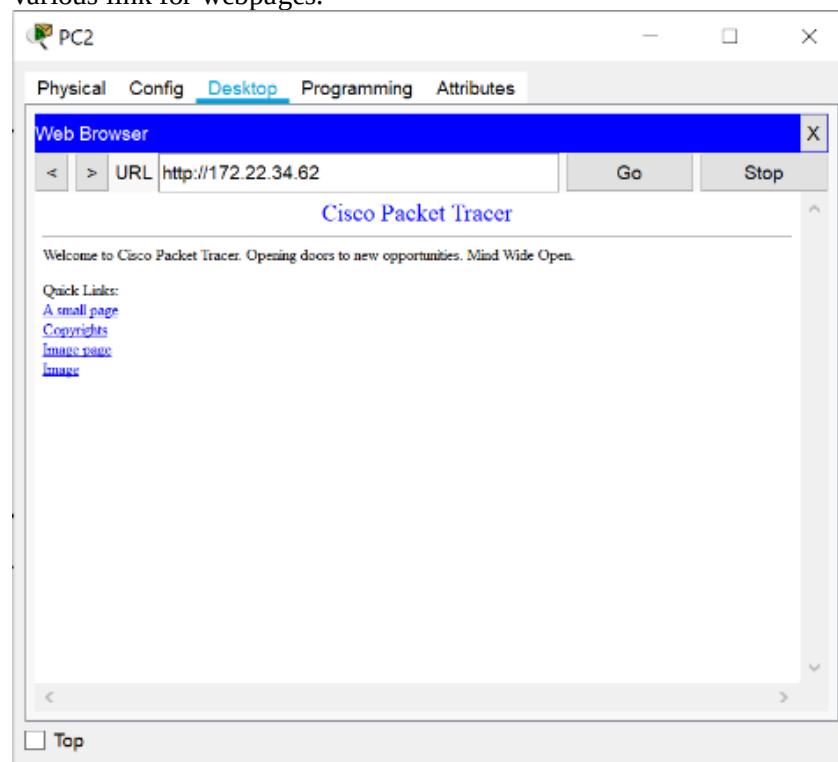
Reply from 172.22.34.62: bytes=32 time=1ms TTL=127
Reply from 172.22.34.62: bytes=32 time=17ms TTL=127
Reply from 172.22.34.62: bytes=32 time<1ms TTL=127
Reply from 172.22.34.62: bytes=32 time<1ms TTL=127

Ping statistics for 172.22.34.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 17ms, Average = 4ms

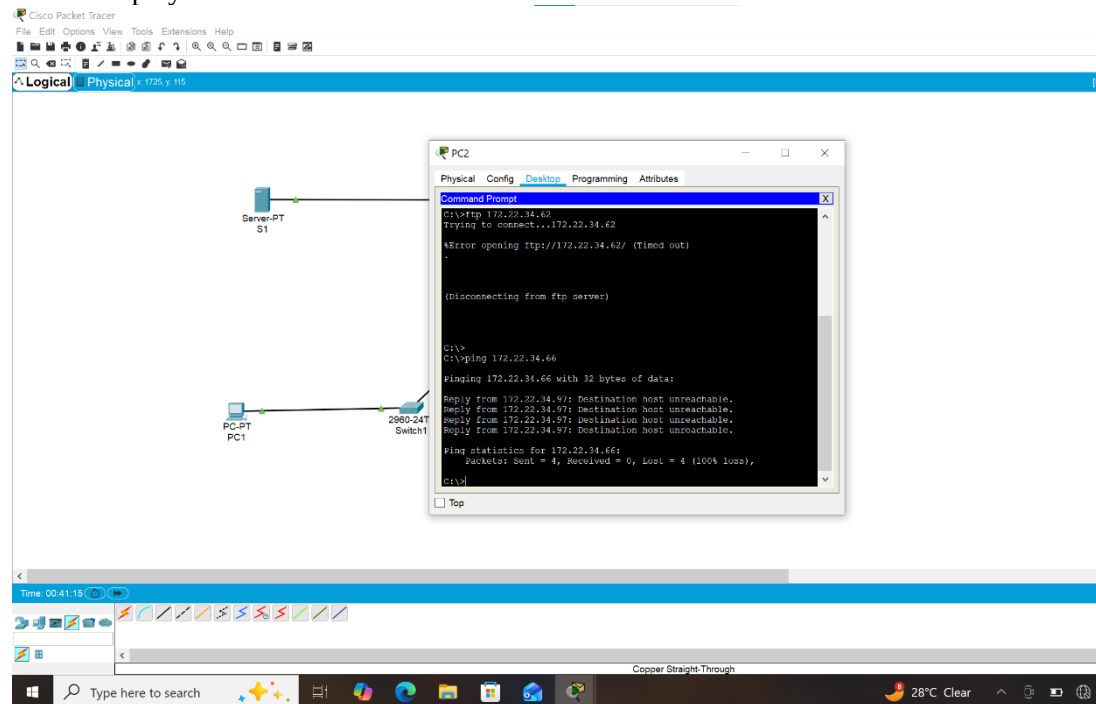
C:\>ftp 172.22.34.62
Trying to connect...172.22.34.62
%Error opening ftp://172.22.34.62/ (Timed out)
.

(Disconnecting from ftp server)
```

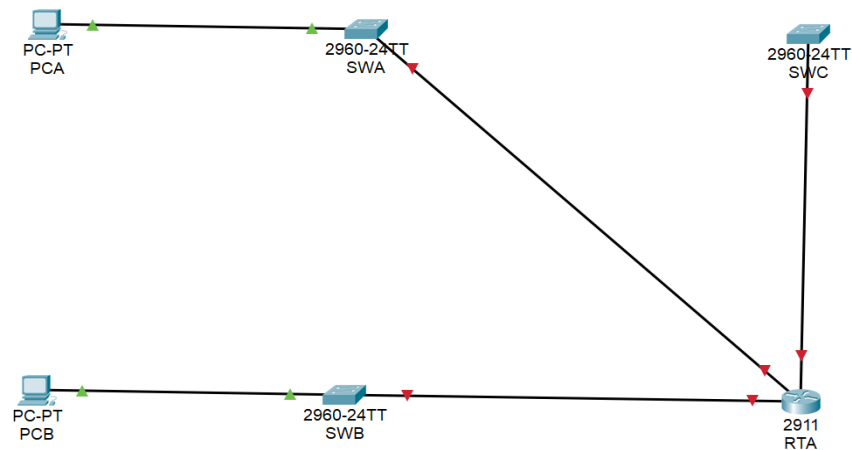
Now go to the web browser in the PC2 and search “ <http://172.22.34.62>” it should display various link for webpages.



Now go to command prompt of PC2 again and try to ping PC1 “ping 172.22.34.66” it should display as destination host unreachable.



Scenario 2:
Topology:



Step 1: Place device according to the topology

Step 2: Making connections using copper straight wire

Step 3: Give IP address, subnet mask, and default gateway(PC) and keep the port status on

PCA

PhysicalConfigDesktopProgrammingAttributes

IP Configuration

InterfaceFastEthernet0

IP Configuration

☐ DHCP

☒ Static

IP Address10.101.117.51

Subnet Mask255.255.255.248

Default Gateway10.101.117.49

DNS Server0.0.0.0

IPv6 Configuration

☐ DHCP

☐ Auto Config

☒ Static

IPv6 Address

Link Local AddressFE80::202:4AFF:FE53:2CD6

IPv6 Gateway

IPv6 DNS Server

802.1X

☐ Use 802.1X Security

AuthenticationMD5

☐ Top

Step 4: Configure the IP address on router

RTA

PhysicalConfigCLIAttributes

GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1

Global Settings

Display NameRTA

HostnameRTA

NVRAM

Erase

Save

Startup Config

Load...

Export...

Running Config

Export...

Merge...

Equivalent IOS Commands

dialog? [yes/no]: n

Press RETURN to get started!

☐ Top



Physical Config CLI Attributes

- Settings
- Algorithm Settings
- ROUTING**
- Static
- RIP
- SWITCHING**
- VLAN Database
- INTERFACE**
- GigabitEthernet0/0
- GigabitEthernet0/1
- GigabitEthernet0/2

GigabitEthernet0/0	
Port Status	
Bandwidth	☐ 1000 Mbps ☒ 100 Mbps ☐ 10 M
Duplex	☐ Half Duplex ☒ Full Du
MAC Address	0002.177B.8001
IP Configuration	
IP Address	10.101.117.49
Subnet Mask	255.255.255.248
Tx Ring Limit	
	10

Equivalent IOS Commands

```
RTA(config-if)#ip address 10.101.117.1 255.255.255.248
RTA(config-if)#ip address 10.101.117.1 255.255.255.224
RTA(config-if)#
RTA(config-if)#exit
RTA(config)#interface GigabitEthernet0/0
RTA(config-if)#
```

☐ Top



Physical Config CLI Attributes

- Settings
- Algorithm Settings
- ROUTING**
- Static
- RIP
- SWITCHING**
- VLAN Database
- INTERFACE**
- GigabitEthernet0/0
- GigabitEthernet0/1
- GigabitEthernet0/2

GigabitEthernet0/1	
Port Status	
Bandwidth	☐ 1000 Mbps ☒ 100 Mbps ☐ 10 M
Duplex	☐ Half Duplex ☒ Full Du
MAC Address	0002.177B.8002
IP Configuration	
IP Address	10.101.117.33
Subnet Mask	255.255.255.240
Tx Ring Limit	
	10

Equivalent IOS Commands

```
RTA(config-if)#exit
RTA(config)#interface GigabitEthernet0/0
RTA(config-if)#
RTA(config-if)#exit
RTA(config)#interface GigabitEthernet0/1
RTA(config-if)#
```

☐ Top



Physical Config CLI Attributes

- Settings
- Algorithm Settings
- ROUTING**
- Static
- RIP
- SWITCHING**
- VLAN Database
- INTERFACE**
- GigabitEthernet0/0
- GigabitEthernet0/1
- GigabitEthernet0/2**

GigabitEthernet0/2

Port Status
Bandwidth ☐ 1000 Mbps ☒ 100 Mbps ☐ 10
Duplex ☐ Half Duplex ☒ Full D
MAC Address 0002.177B.8003
IP Configuration
IP Address 10.101.117.1
Subnet Mask 255.255.255.224
Tx Ring Limit 10

Equivalent IOS Commands

```
RTA(config-if)#exit  
RTA(config)#interface GigabitEthernet0/1  
RTA(config-if)#  
RTA(config-if)#exit  
RTA(config)#interface GigabitEthernet0/2  
RTA(config-if)#
```

☐ Top

Give IP address of the switch



Physical Config CLI Attributes

IOS Command Line Interface

```
Switch#configure terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
Switch(config)#hostname SWA  
SWA(config)#  
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to u  
  
SWA(config)#  
SWA(config)#int vlan 1  
SWA(config-if)#ip address 10.101.117.50 255.255.255.248  
SWA(config-if)#no shut  
  
SWA(config-if)#  
%LINK-5-CHANGED: Interface Vlan1, changed state to up  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up  
  
SWA(config-if)#ip default-gateway 10.101.117.49  
SWA(config)#
```

Ctrl+F6 to exit CLI focus

Copy

☐ Top

 SWB

Physical Config CLI Attributes

IOS Command Line Interface

```
state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/1, changed state to up

SWB(config)#int vlan 1
SWB(config-if)#ip address 10.101.117.34
255.255.255.240
SWB(config-if)#no shut

SWB(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to
up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
Vlan1, changed state to up

SWB(config-if)#ip default-gateway 10.101.117.33
SWB(config)#
SWB(config)#
```

Ctrl+F6 to exit CLI focus

Copy

Pas

☐ Top

 SWC

Physical Config CLI Attributes

IOS Command Line Interface

```
SWC>en
SWC#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SWC(config)#int vlan 1
SWC(config-if)#ip address 10.101.117.2 255.255.255.224
SWC(config-if)#no shut

SWC(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed sta
to up

SWC(config-if)#ip default-gateway 10.101.117.1
SWC(config)#
SWC(config)#
```

Ctrl+F6 to exit CLI focus

Copy

P

☐ Top

Configure secret and console password on router and switch

```
SWA(config)#ip default-gateway 10.101.117.254
SWA(config)#
SWA(config)#enable secret enpa55
SWA(config)#line console 0
SWA(config-line)#password tyit
SWA(config-line)#login
SWA(config-line)#
SWA(config-line)#
```

```
SWB(config)#enable secret enpa55
SWB(config)#line console 0
SWB(config-line)#password tyit
SWB(config-line)#login
SWB(config-line)#
SWB(config-line)#
```

```
SWC(config-line)#
SWC(config-line)#enable secret enpa55
SWC(config)#line console 0
SWC(config-line)#password tyit
SWC(config-line)#login
```

```
SWC(config-line)#
```

```
RTA>en
```

```
RTA#configure terminal
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
RTA(config)#enable secret enpa55
```

```
RTA(config)#line console 0
```

```
RTA(config-line)#password tyit
```

```
RTA(config-line)#login
```

```
RTA(config-line)#
```

```
RTA(config-line)#
```

Check Connectivity between PCA and PCB as well as PCA and PCB to SWC

```
C:\>ping 10.101.117.35
```

```
Pinging 10.101.117.35 with 32 bytes of data:
```

```
Reply from 10.101.117.35: bytes=32 time=1ms TTL=127
Reply from 10.101.117.35: bytes=32 time=12ms TTL=127
Reply from 10.101.117.35: bytes=32 time=2ms TTL=127
Reply from 10.101.117.35: bytes=32 time<1ms TTL=127
```

```
Ping statistics for 10.101.117.35:
```

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 3ms
```

```
C:\>ping 10.101.117.2
```

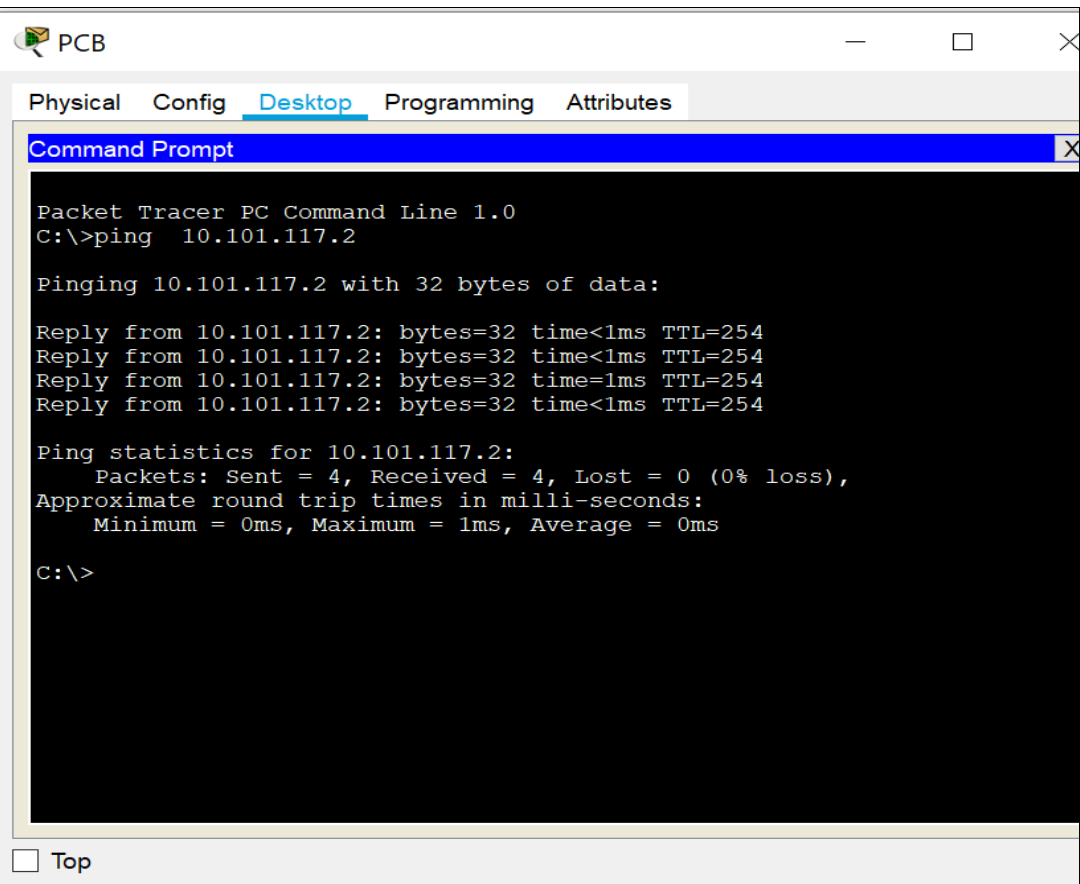
```
Pinging 10.101.117.2 with 32 bytes of data:
```

```
Reply from 10.101.117.2: bytes=32 time=1ms TTL=254
Reply from 10.101.117.2: bytes=32 time=4ms TTL=254
Reply from 10.101.117.2: bytes=32 time=1ms TTL=254
Reply from 10.101.117.2: bytes=32 time<1ms TTL=254
```

```
Ping statistics for 10.101.117.2:
```

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 1ms
```

```
C:\>
```



PCB

Physical Config Desktop Programming Attributes

Command Prompt

```
Packet Tracer PC Command Line 1.0
C:\>ping 10.101.117.2

Pinging 10.101.117.2 with 32 bytes of data:

Reply from 10.101.117.2: bytes=32 time<1ms TTL=254
Reply from 10.101.117.2: bytes=32 time<1ms TTL=254
Reply from 10.101.117.2: bytes=32 time=1ms TTL=254
Reply from 10.101.117.2: bytes=32 time<1ms TTL=254

Ping statistics for 10.101.117.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

☐ Top

Configure the domain name, username and vty lines and crypto key on routers and switches

```
SWA(config-line)#
SWA(config-line)#exit
SWA(config)#ip domain-name ccnasecurity.com
SWA(config)#line vty 04
SWA(config-line)#login local
SWA(config-line)#crypto key generate rsa
The name for the keys will be: SWA.ccnasecurity.com
Choose the size of the key modulus in the range of
360 to 2048 for your
  General Purpose Keys. Choosing a key modulus
greater than 512 may take
  a few minutes.
```

```
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-
exportable...[OK]
```

```
SWA(config)#
```

```
SWB(config-line)#exit
SWB(config)#ip domain-name ccnasecurity.com
SWB(config)#username admin secret adminpa55
SWB(config)#line vty 04
SWB(config-line)#login local
SWB(config-line)#crypto key generate rsa
The name for the keys will be: SWB.ccnasecurity.com
Choose the size of the key modulus in the range of
360 to 2048 for your
  General Purpose Keys. Choosing a key modulus
greater than 512 may take
  a few minutes.
```

```
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-
exportable...[OK]
```

```
SWB(config)#
```



Physical Config CLI Attributes

IOS Command Line Interface

```
SWC#configure terminal
Enter configuration commands, one per line. End
with CNTL/Z.
SWC(config)#ip domain-name ccnasecurity.com
SWC(config)#username admin secret adminpa55
SWC(config)#line vty 04
SWC(config-line)#login local
SWC(config-line)#crypto key generate rsa
The name for the keys will be: SWC.ccnasecurity.com
Choose the size of the key modulus in the range of
360 to 2048 for your
  General Purpose Keys. Choosing a key modulus
greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-
exportable...[OK]

SWC(config)#
```

Ctrl+F6 to exit CLI focus

Copy

Paste

☐ Top



Physical Config CLI Attributes

IOS Command Line Interface

```
Password:
RTA#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
RTA(config)#username admin secret adminpa55
RTA(config)#ip domain-name ccnasecurity.com
RTA(config)#username admin secret adminpa55
RTA(config)#line vty 04
RTA(config-line)#login local
RTA(config-line)#crypto key generate rsa
The name for the keys will be: RTA.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

RTA(config)#
*Mar 1 0:43:5.644: %SSH-5-ENABLED: SSH 1.99 has been enabled
RTA(config)#
```

Ctrl+F6 to exit CLI focus

Copy

Paste

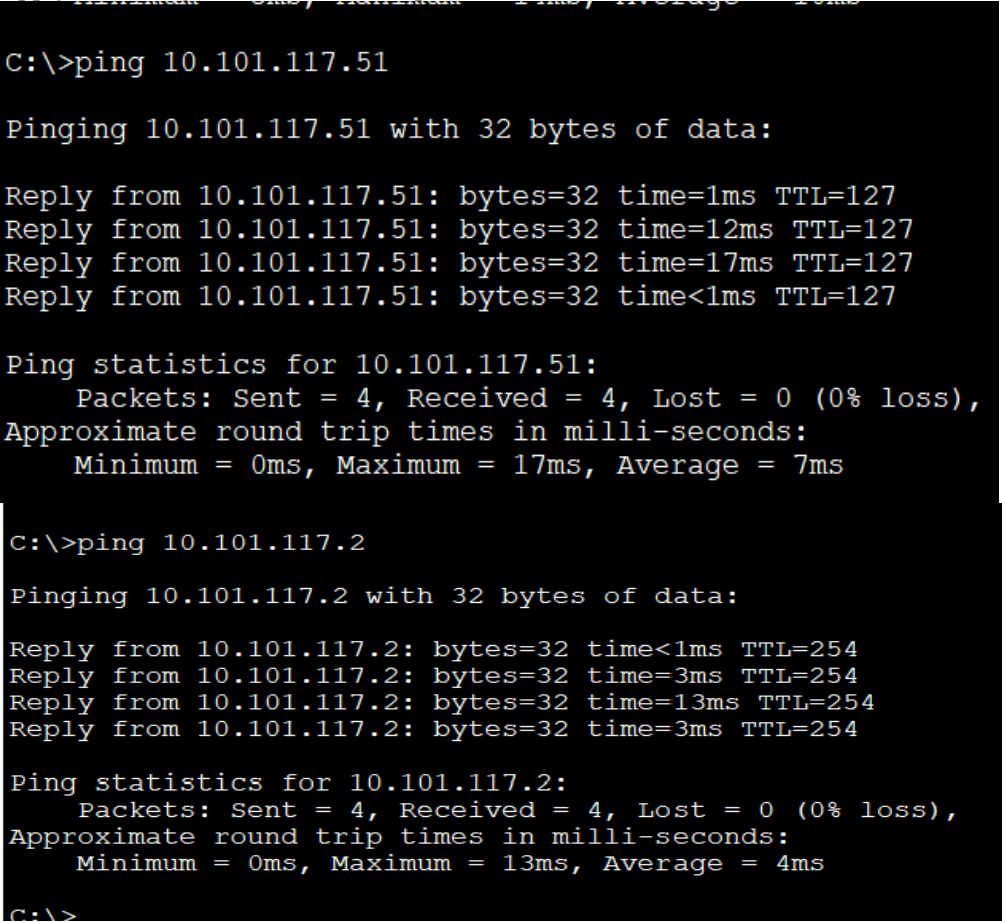
☐ Top

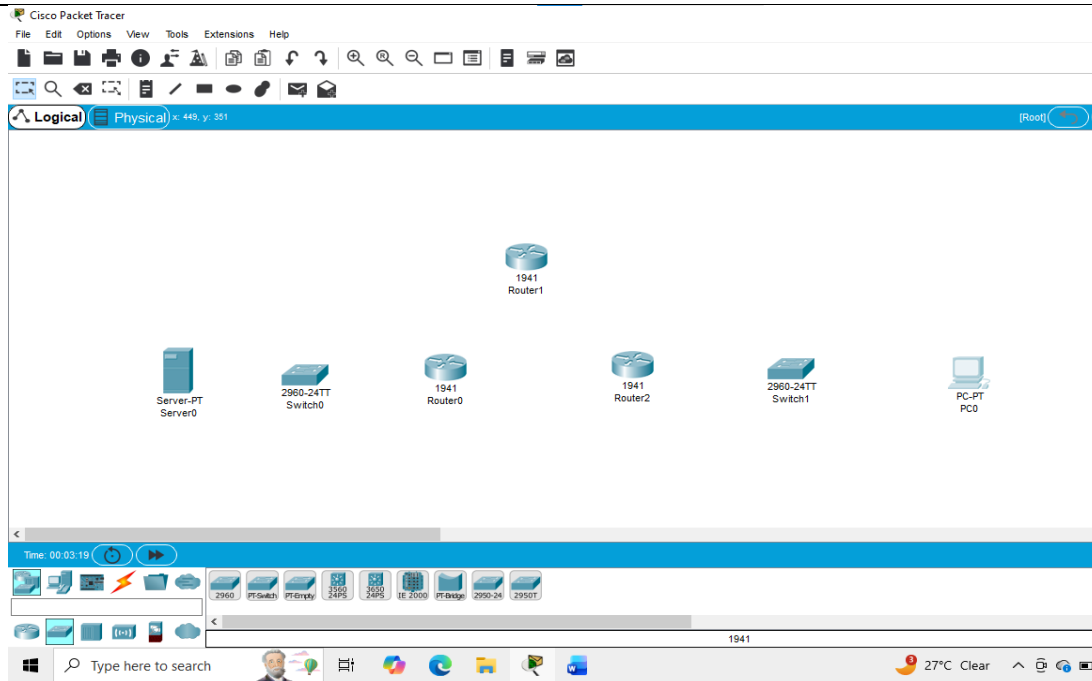
Verify the SSH connections using the below given command

```
C:\>ssh -l Admin 10.101.117.34
```

Password:

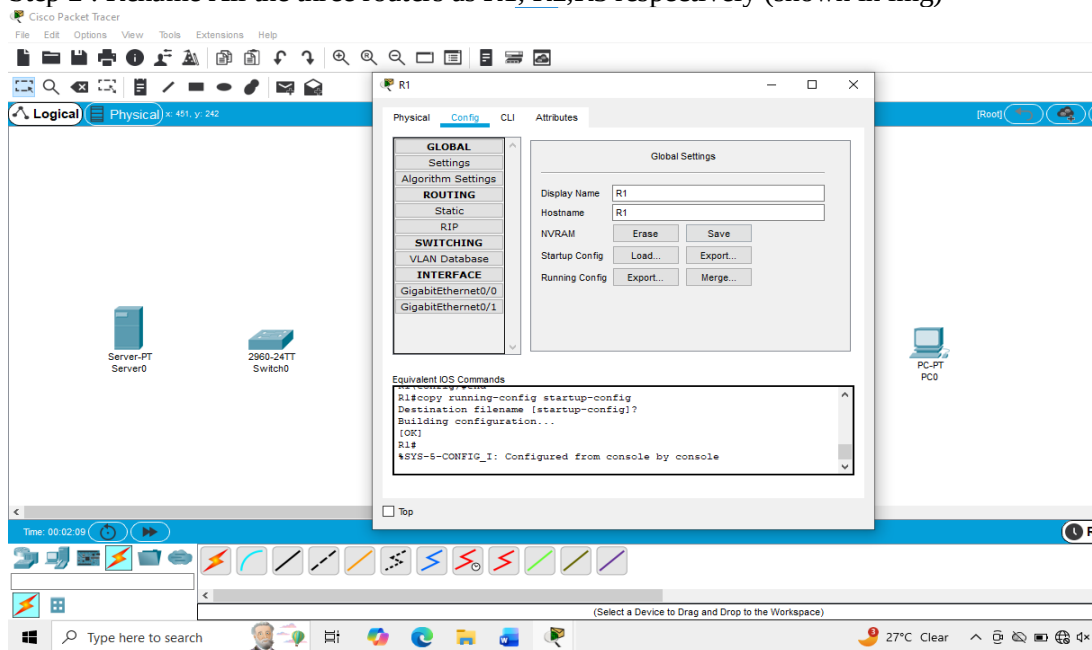
Configure extended ACL on the router

	<pre> User Access Verification Password: RTA>en Password: RTA#configure terminal Enter configuration commands, one per line. End with CNTL/Z. RTA(config)#access-list 199 permit tcp 10.101.117.32 0.0.0.15 10.101.117.0 0.0.0.31 eq 22 RTA(config)#access-list 199 permit icmp any any RTA(config)#int gig0/2 RTA(config-if)#ip access-group 199 out RTA(config-if)# </pre> Apply and verify extended ACL implementation  <pre> C:\>ping 10.101.117.51 Pinging 10.101.117.51 with 32 bytes of data: Reply from 10.101.117.51: bytes=32 time=1ms TTL=127 Reply from 10.101.117.51: bytes=32 time=12ms TTL=127 Reply from 10.101.117.51: bytes=32 time=17ms TTL=127 Reply from 10.101.117.51: bytes=32 time<1ms TTL=127 Ping statistics for 10.101.117.51: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 17ms, Average = 7ms C:\>ping 10.101.117.2 Pinging 10.101.117.2 with 32 bytes of data: Reply from 10.101.117.2: bytes=32 time<1ms TTL=254 Reply from 10.101.117.2: bytes=32 time=3ms TTL=254 Reply from 10.101.117.2: bytes=32 time=13ms TTL=254 Reply from 10.101.117.2: bytes=32 time=3ms TTL=254 Ping statistics for 10.101.117.2: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 0ms, Maximum = 13ms, Average = 4ms C:\> </pre>
Practical 4	Practical 4 4.A Verify connectivity among devices before firewall configuration. 2-2-26 Step 1 : open cisco packet tracer And setup an topology shown in the image

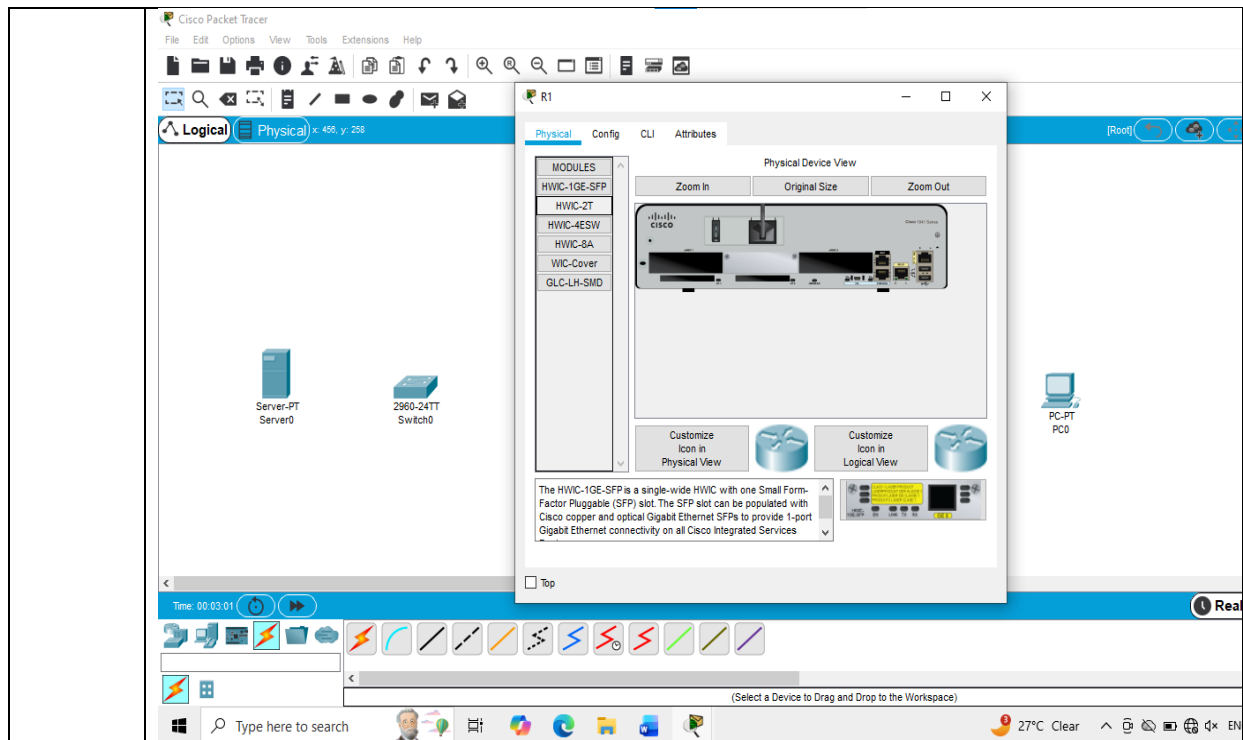


Appratuused : Router (1941) x 3
 Switch (2960) x 2
 PC-PT x 1
 Server-PT x 1

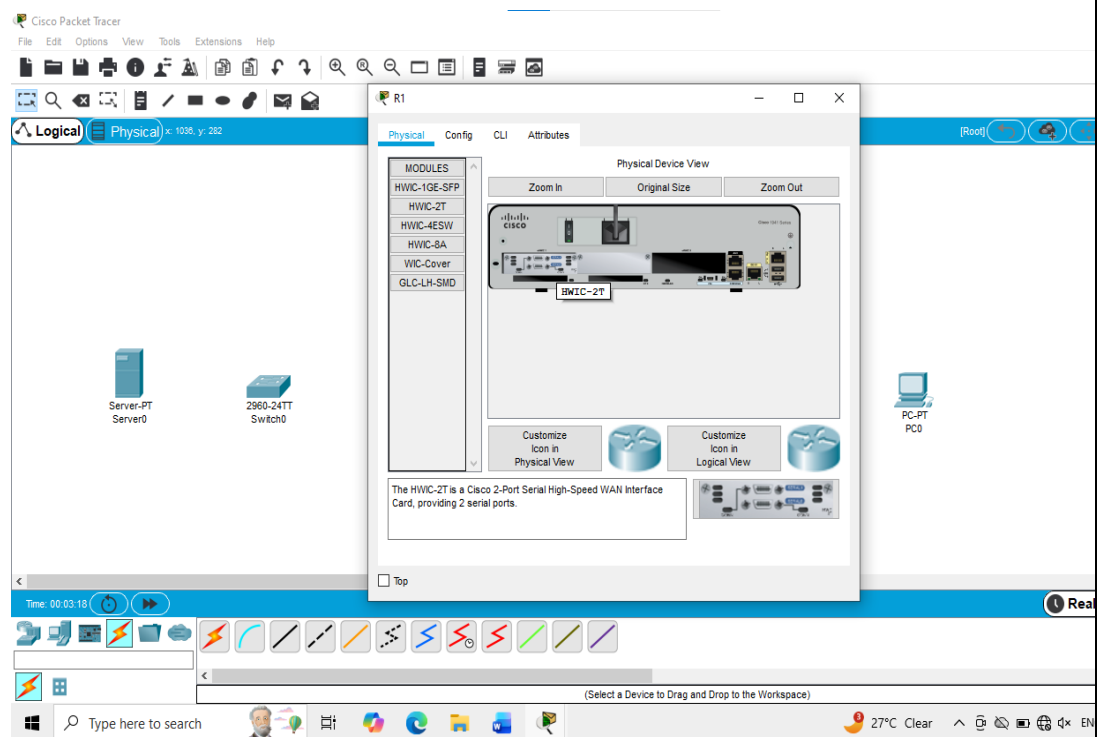
Step 2 : Rename All the three routers as R1, R2, R3 respectively (shown in img)



Step 3: Go to router R1 -> physical -> switch OFF the machine

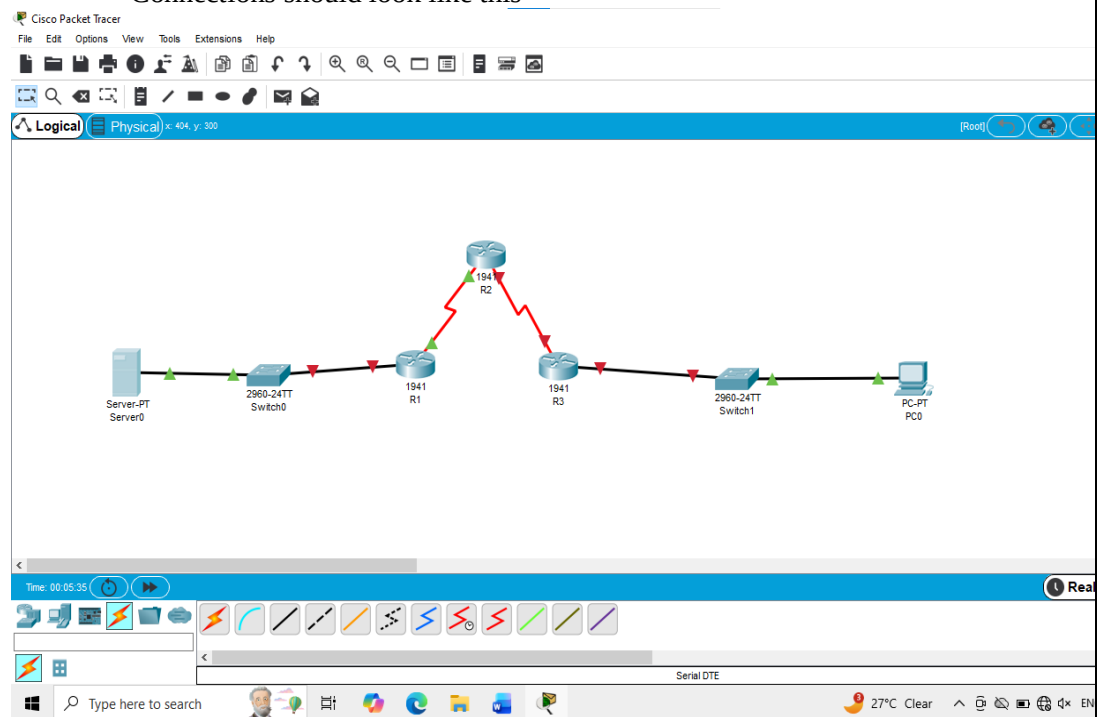


- Select HWIC-2T from modules and drag and drop it



- Turn the machine ON

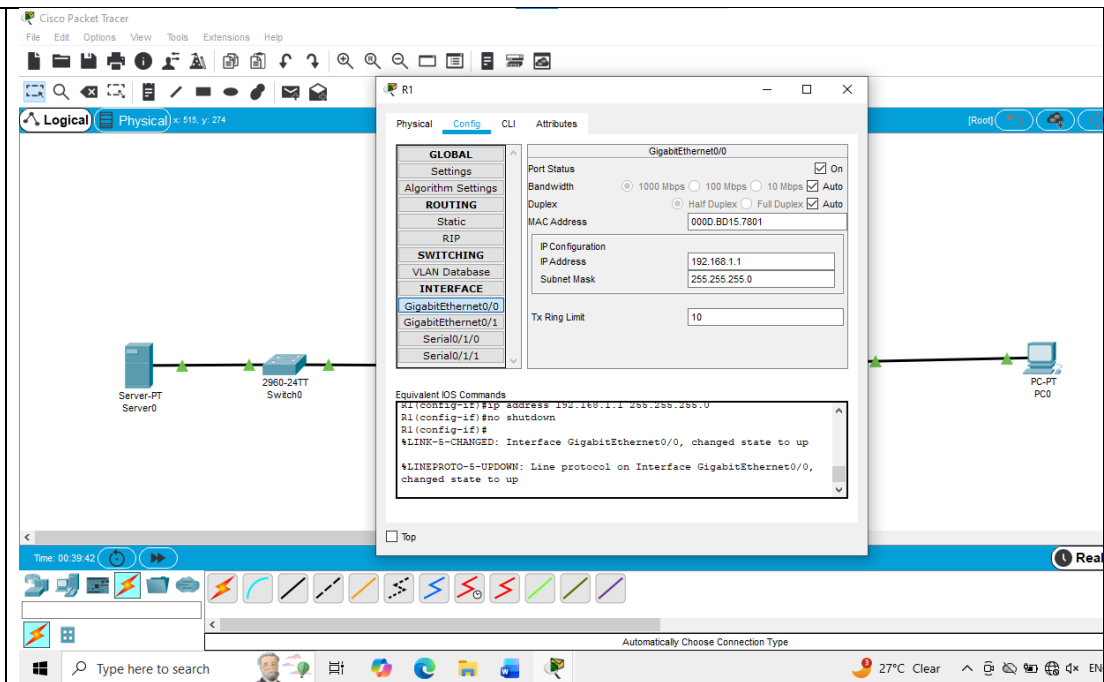
Step 4 : Make connections using the common wire
Connections should look like this



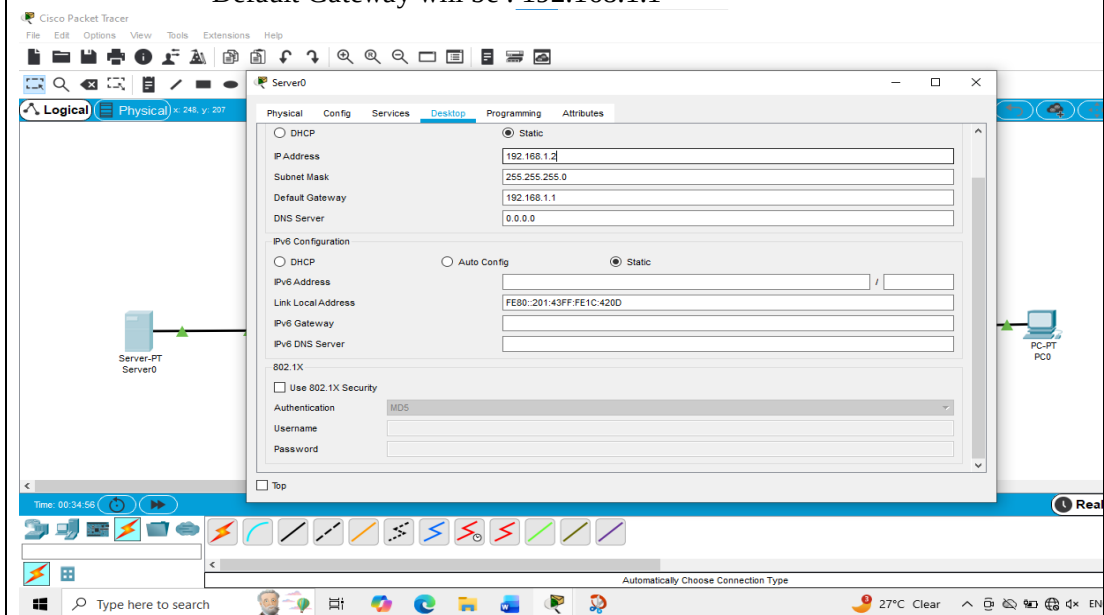
Configuring Routers and PC :

Step 5 : Go To R1 -> Config

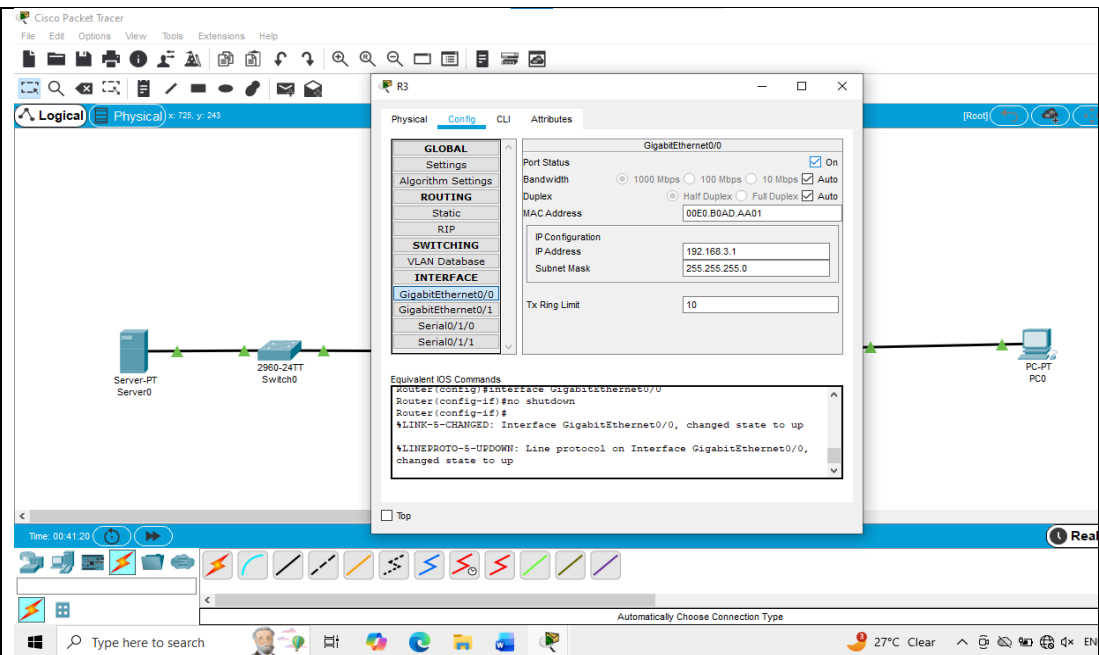
In config locate to GigabitEthernet 0/0 and put IP as 192.168.1.1
And turn it ON



Go to server -> desktop -> IP configuration
 Put IP as :192.168.1.2
 Subnet mask will be : 255.255.25.0
 Default Gateway will be : 192.168.1.1

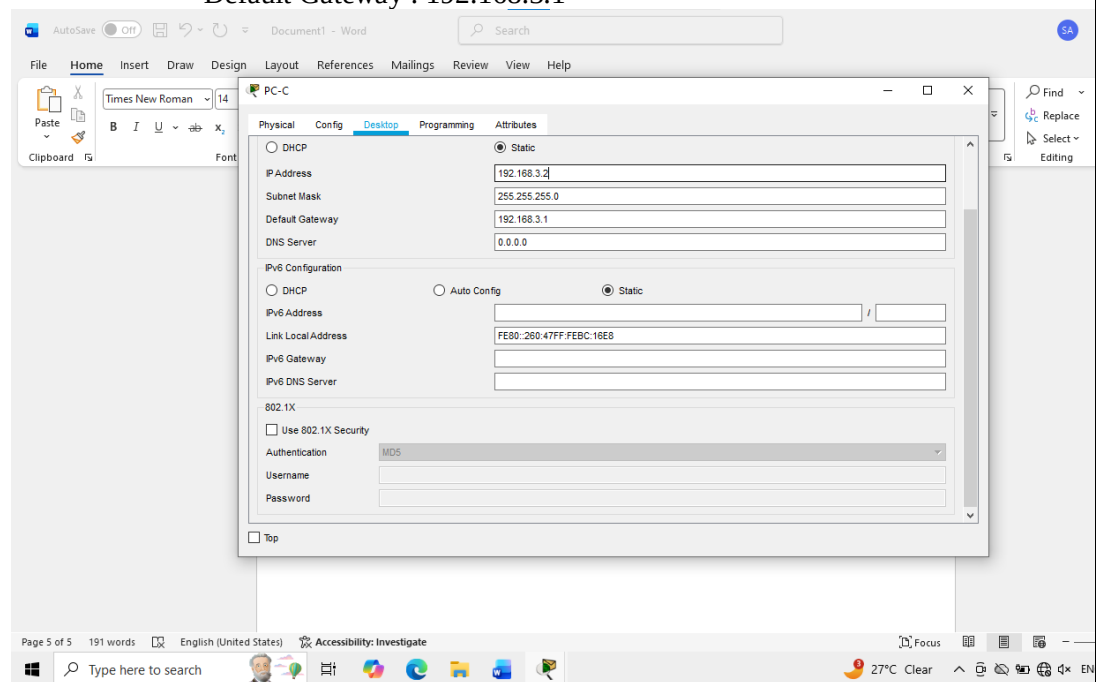


Step 6 : Go to R3 -> Config
 In config locate to GigabitEthernet 0/0 and put IP as 192.168.3.1
 And turn it ON



- Go to PC-C -> Desktop -> IP Configuration

Put IP as : 192.168.3.2
 Subnet mask : 255.255.255.0
 Default Gateway : 192.168.3.1



Step 7 : Configuring Interface between Routers

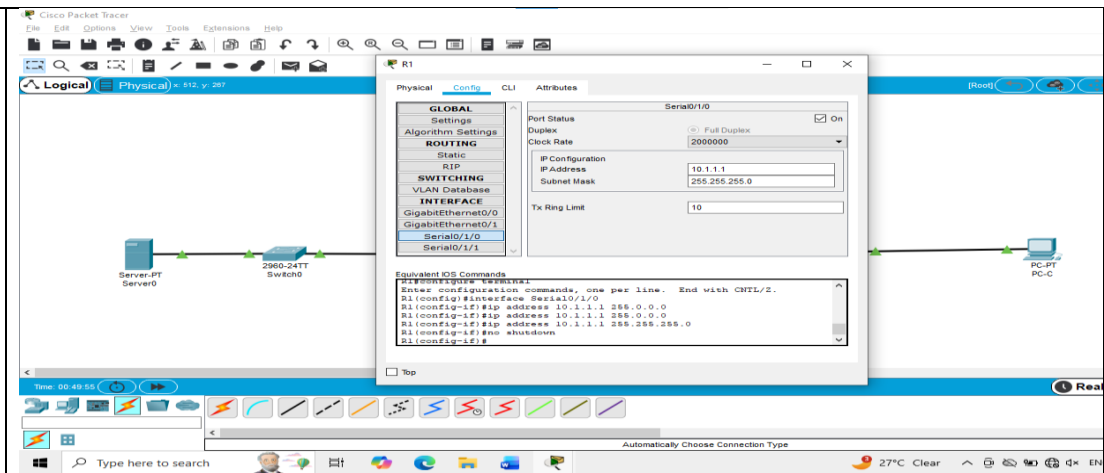
(Before Implementing this first check Serial number / interface on both sides make a note of it)

Go to R1-> Config-> serial 0/1/0

IP : 10.1.1.1

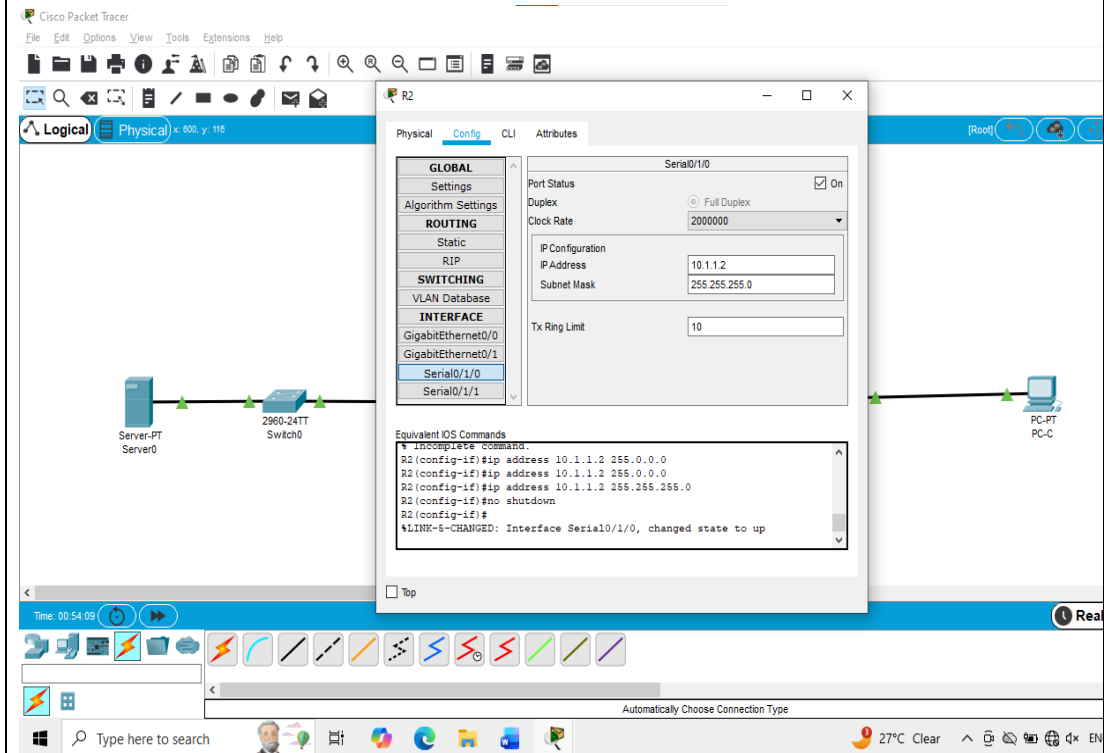
Subnet Mask 255.255.255.0 (Manually change it)

Turn It ON



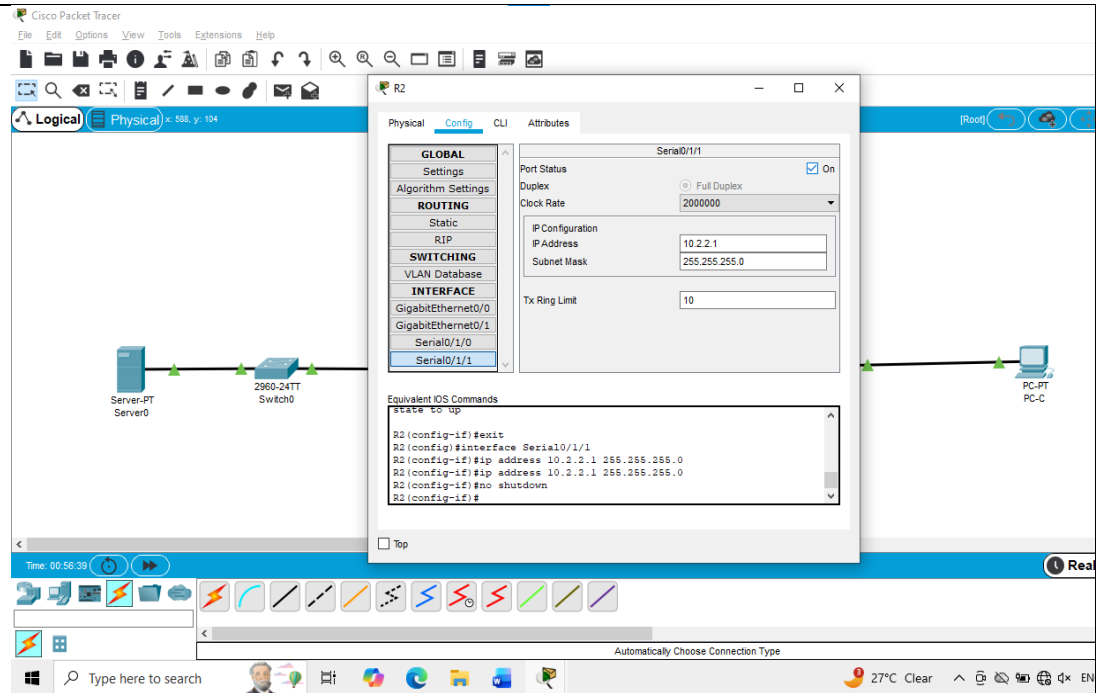
- Go to R2 -> config -> serial 0/1/0

Give IP : 10.1.1.2
Default Gateway : 255.255.255.0
Turn It ON



- Go to R2-> config -> serial 0/1/1

IP : 10.2.2.1
Subnet mask : 255.255.255.0
Turn It ON

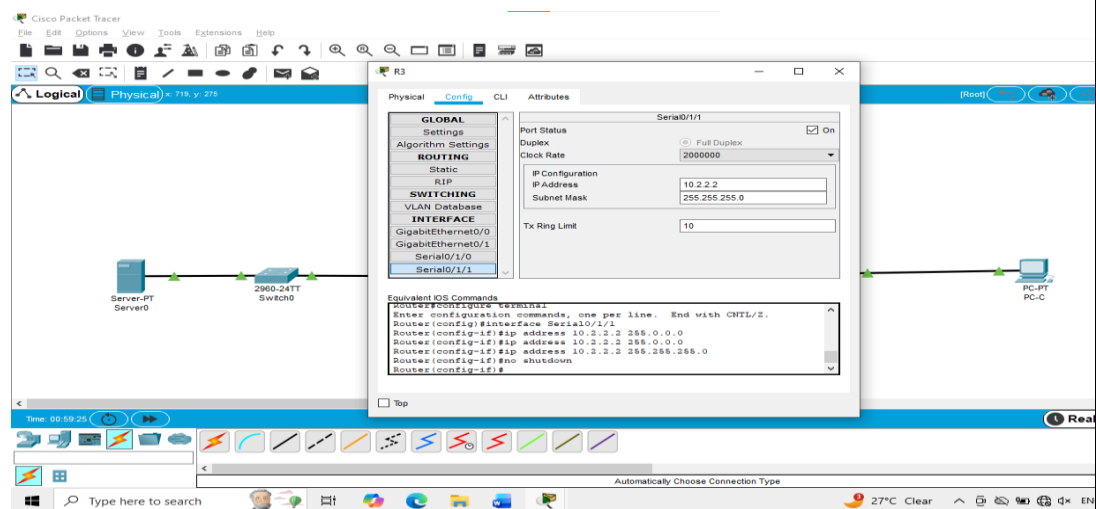


Providing Interface Between R2 and R3

Go to R3-> Config -> serial 0/1/0

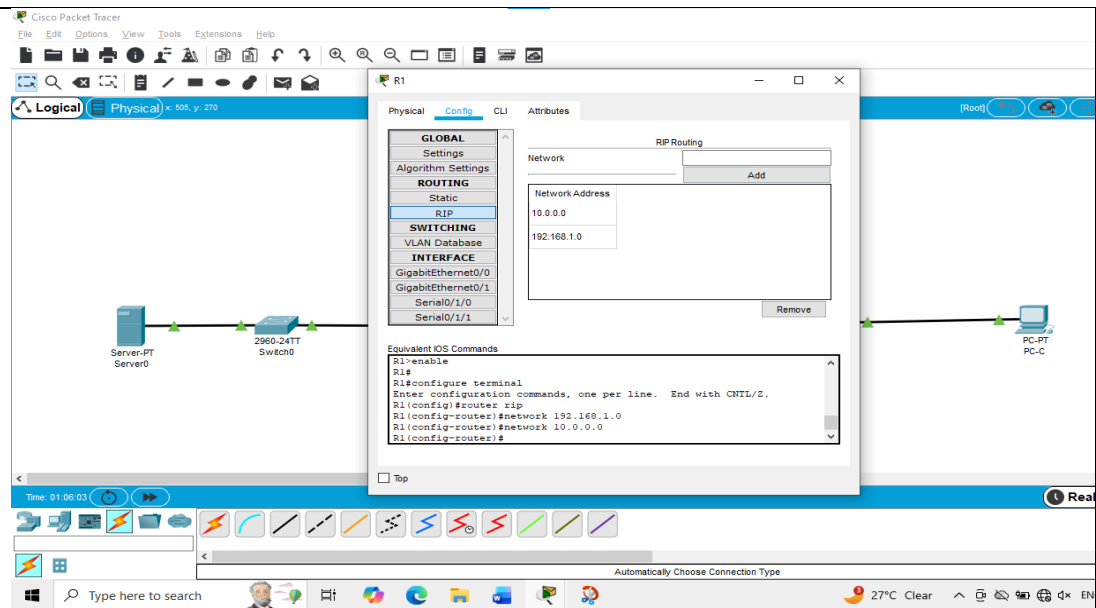
IP : 10.2.2.2

Subnet mask : 255.255.255.0



Setting LoopBack Address For Router R2

- For Implementing this we have to enable the routing scheme first
- Go to R1 -> Config -> find RIP under ROUTING
Add the following IP addresses :
 - 1) 192.168.1.0
 - 2) 10.1.1.0

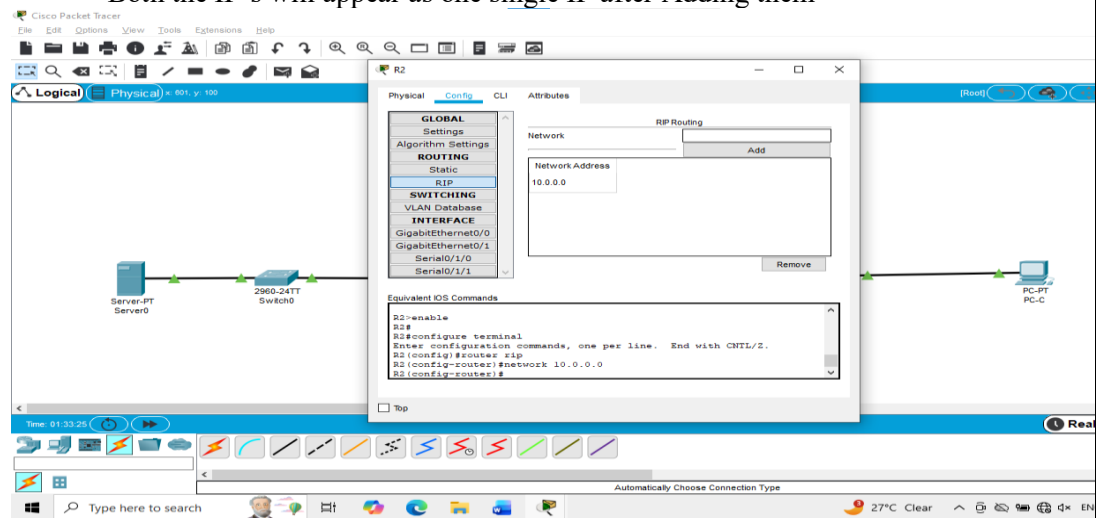


- Similarly do for R2

R2 -> Config -> Locate RIP

Enter IP's : 10.1.1.0 and 10.2.2.0

Both the IP's will appear as one single IP after Adding them



To setup Loopback Address

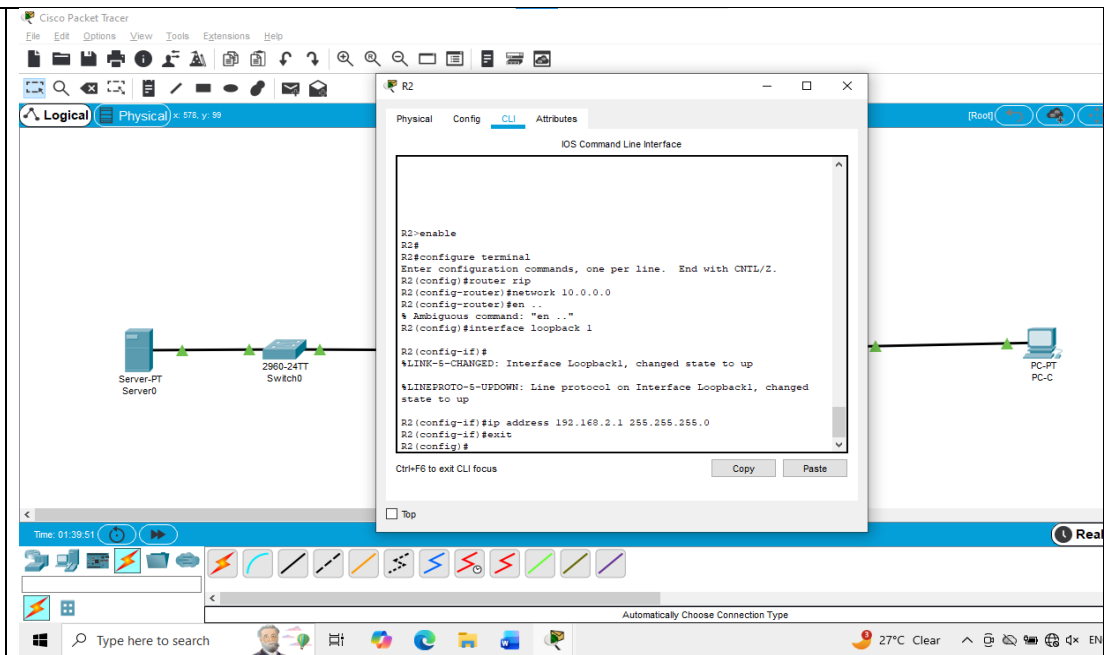
- Go to R2 -> CLI
TYPE commands : en ..

R2(config)#interface loopback 1

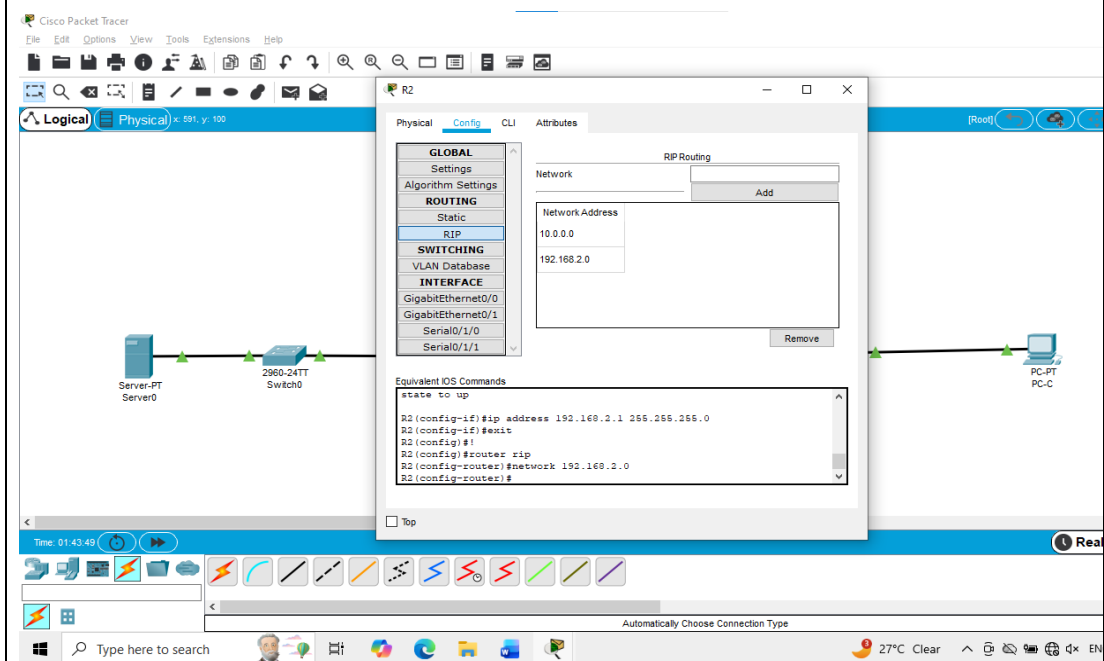
R2(config-if)#ip address 192.168.2.1 255.255.255.0

R2(config-if)#exit

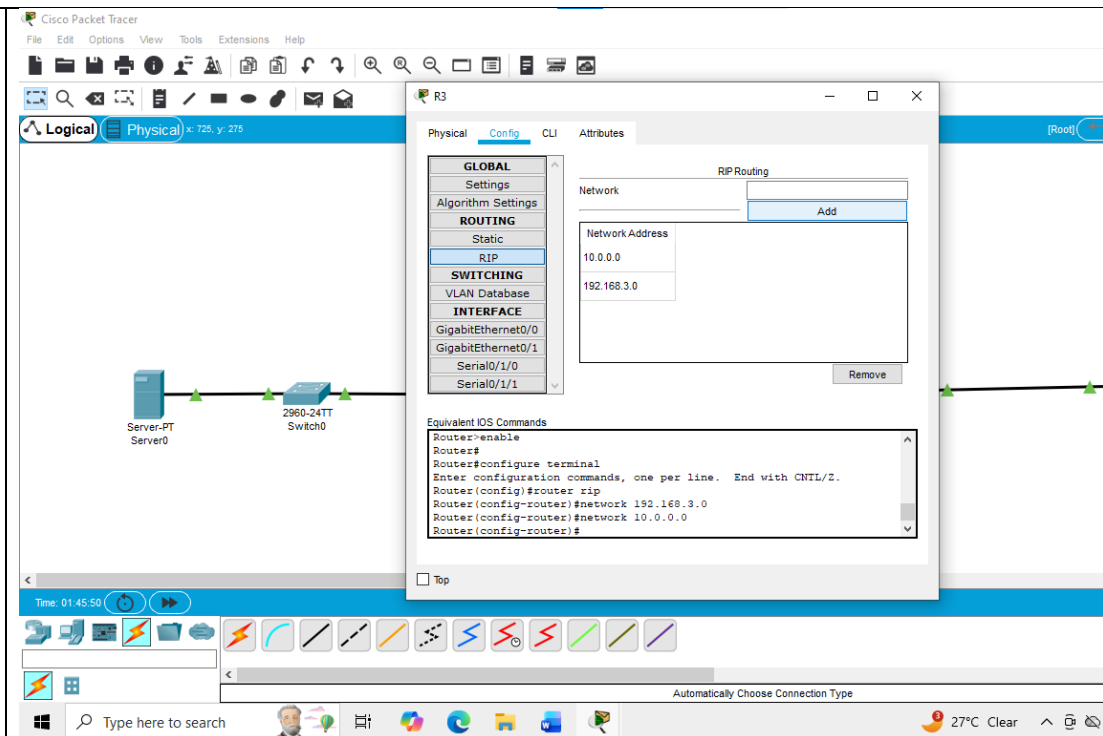
After executing this it should look like this



- Again go to :
R2 -> Config -> Locate RIP
Add IP : 192.168.2.0



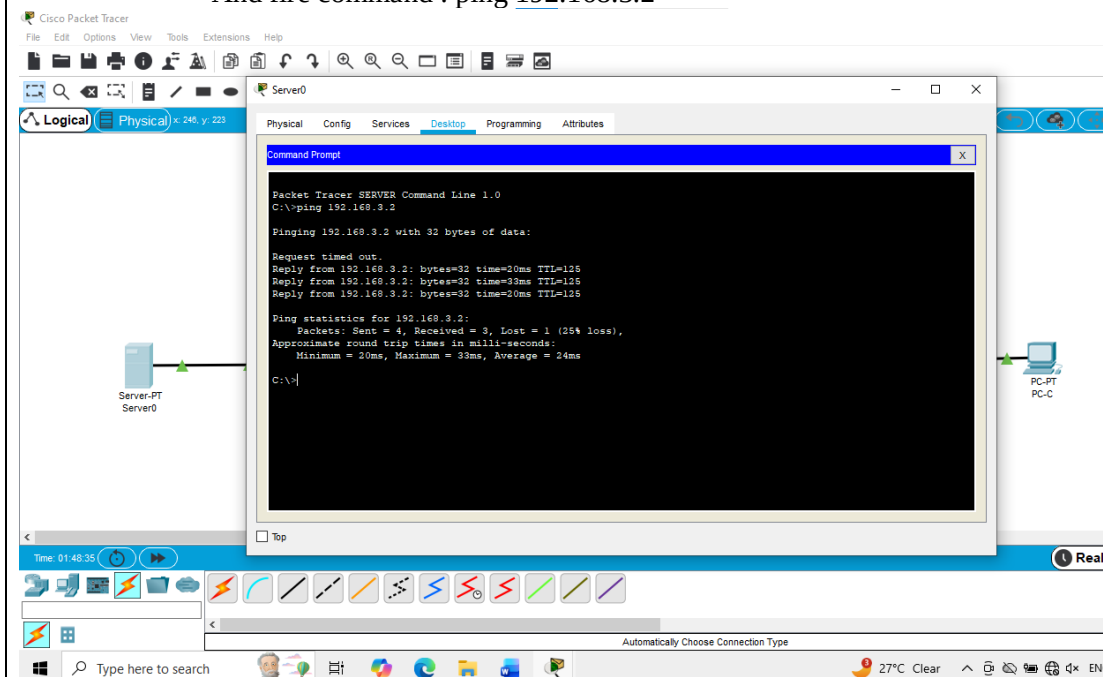
- Go To R3 -> RIP
Add IP addresses : 192.168.3.0 and 10.2.2.0



Step 8 : To check the Connectivity

Go to server -> Desktop -> Command Prompt

And fire command : ping 192.168.3.2



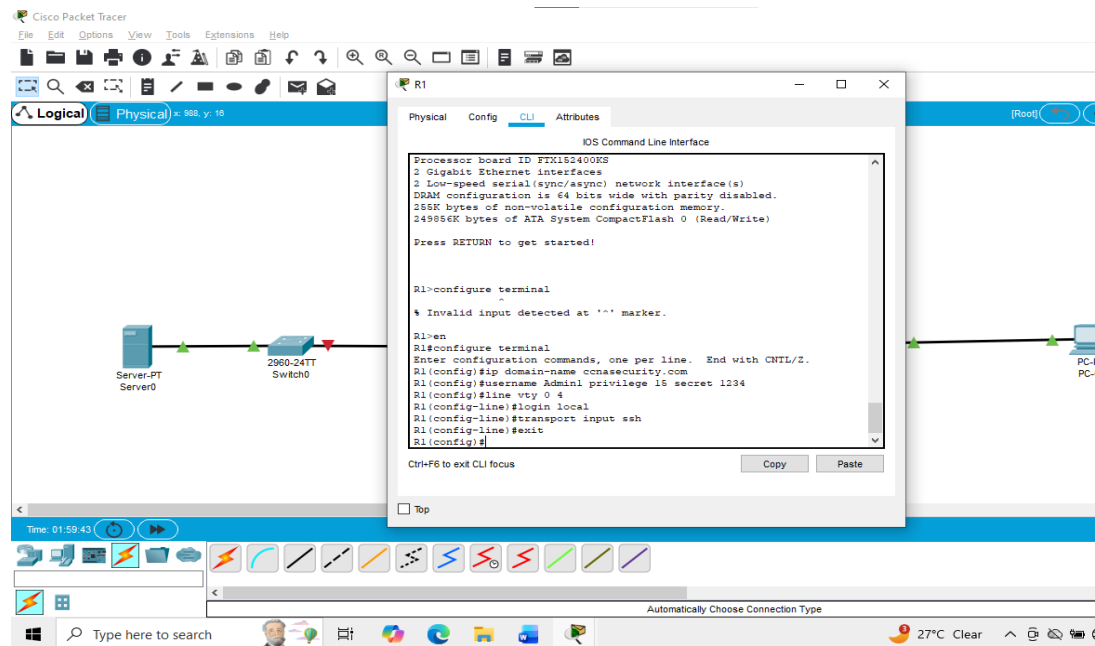
We're getting the reply that means the connection is successful.

Step 9 : To implement SSH protocol

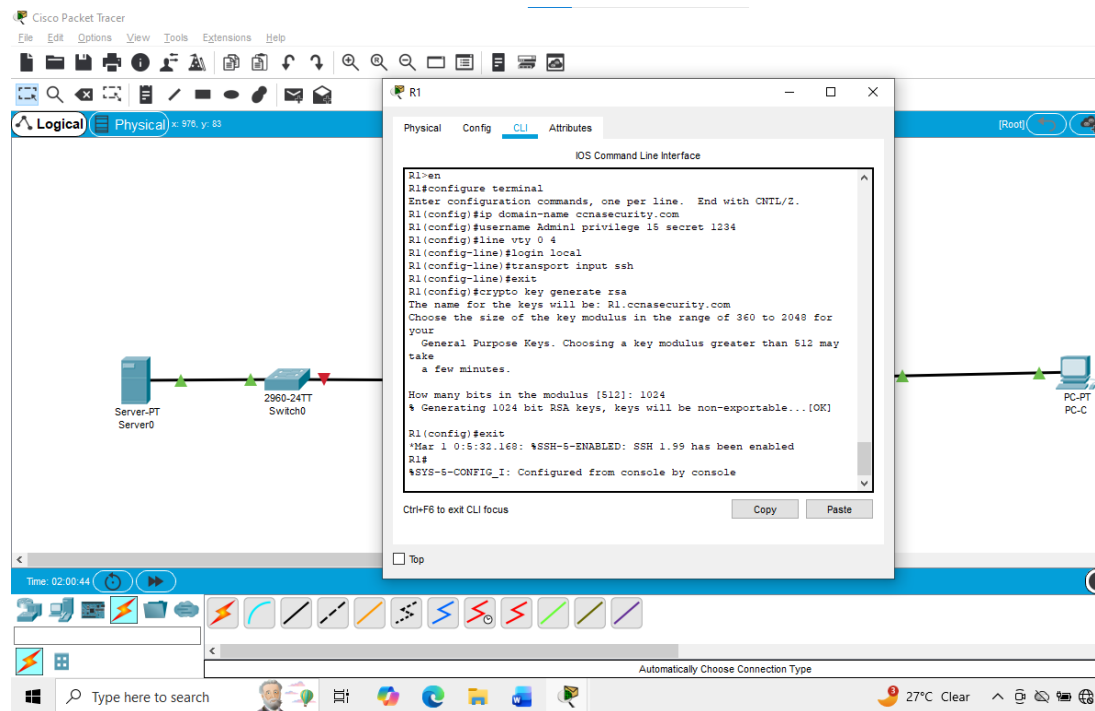
Go to R1 -> CLI -> and enter the following set of commands

- R1(config)#ip domain-name ccnasecurity.com
- R1(config)#username Admin1 privilege 15 secret 1234
- R1(config)#line vty 0 4
- R1(config-line)#login local

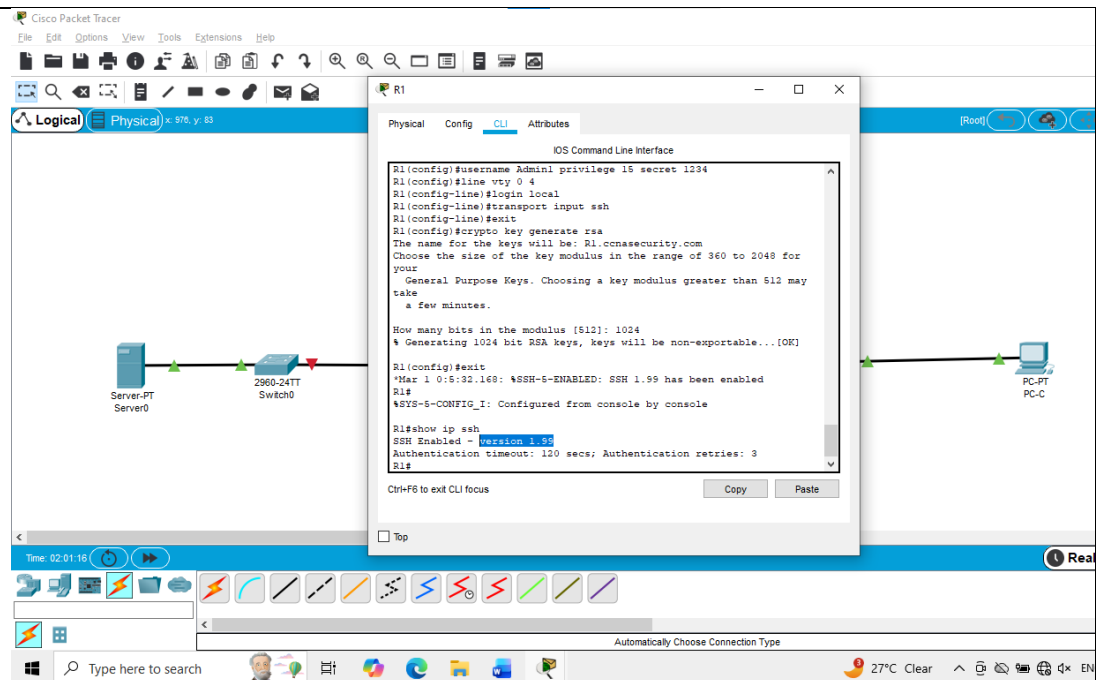
- R1(config-line)#transport input ssh
- R1(config-line)#exit



- SSH has been set Up and to set up a crypto key use
 - R1(config)#crypto key generate rsa
 - It will ask for how many bites give it as 1024



- To check if the SSH configured or not use
 - show ip ssh (Command)



In output it will display : SSH 1.99 has been enabled

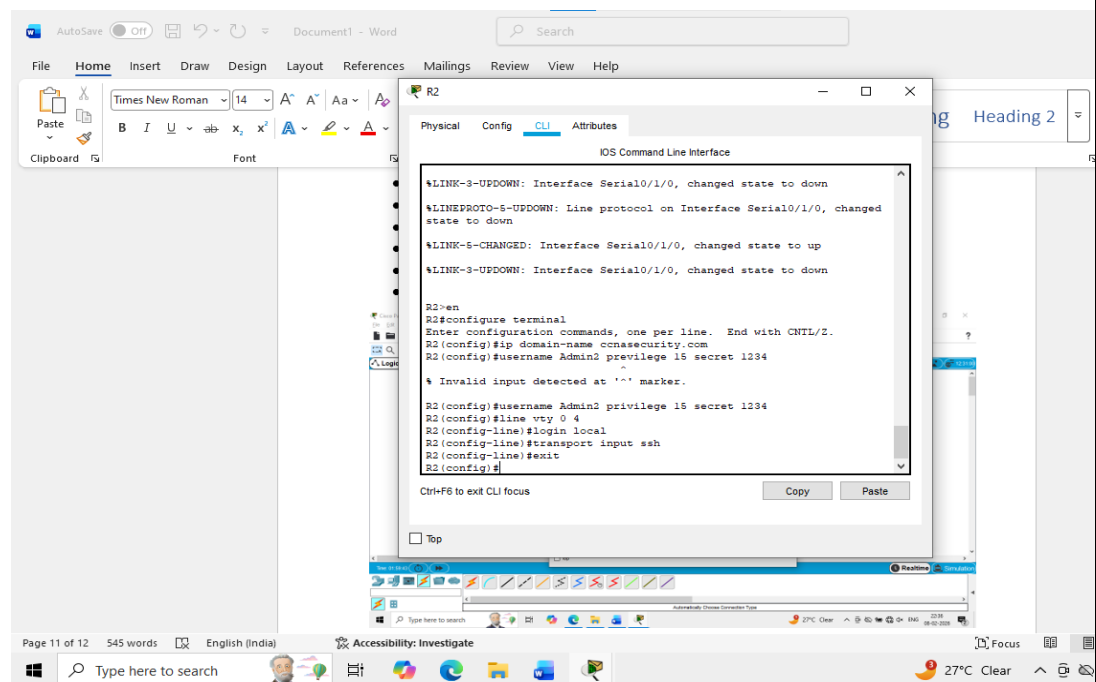
Means SSH has been configured successfully for R1

4.B Use ACLs to ensure remote access to the routers is available only from Management station PC-C.

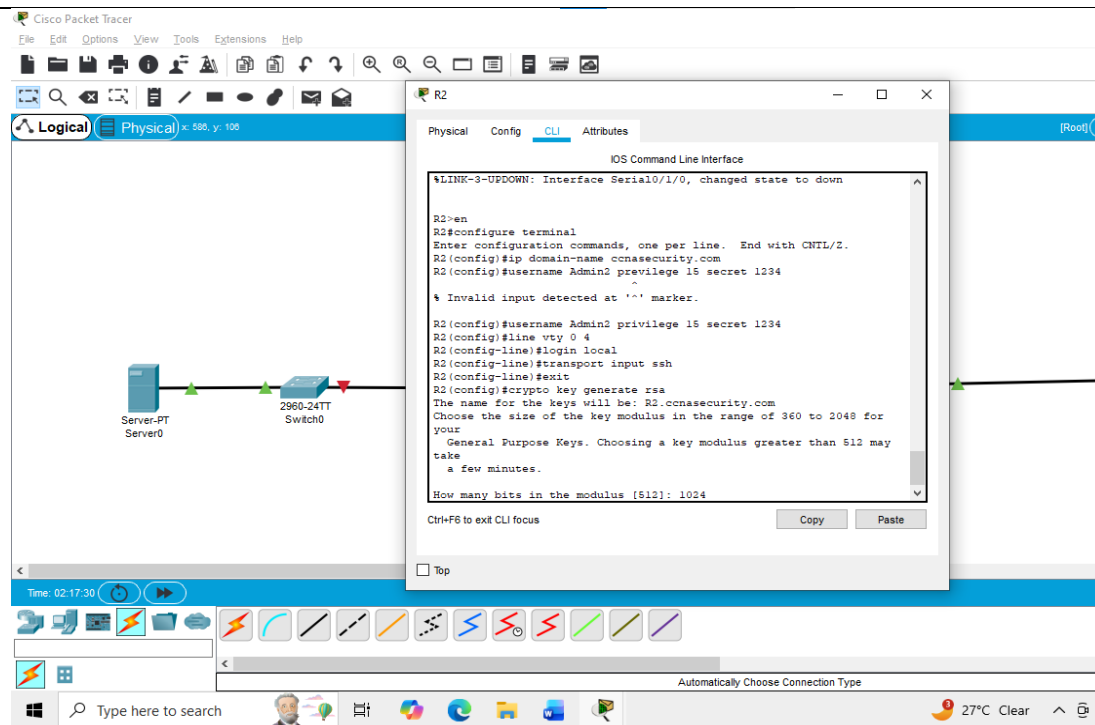
2-2-26

For R2

- similarly use same set of commands for R2 and R3
 - R2(config)#ip domain-name ccnasecurity.com
 - R2(config)#username Admin2 privilege 15 secret 1234
 - R2(config)#line vty 0 4
 - R2(config-line)#login local
 - R2(config-line)#transport input ssh
 - R2(config-line)#exit

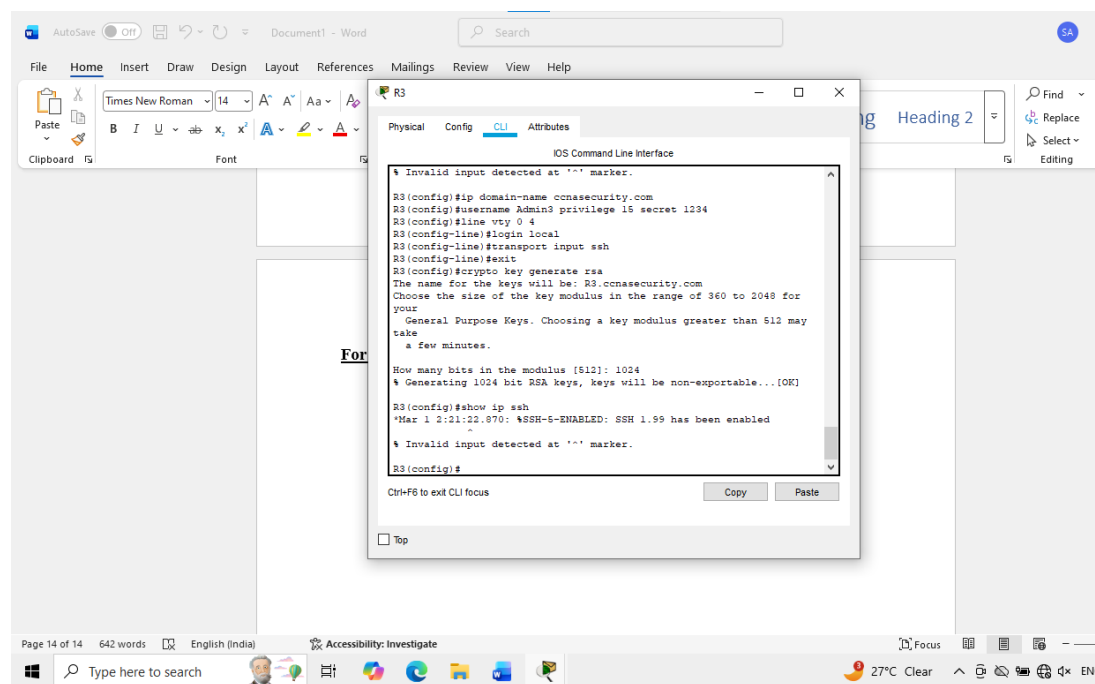


- And also use : R2(config)#crypto key generate rsa (command)



For R3 :

- R3(config)#ip domain-name ccnasecurity.com
- R3(config)#username Admin3 privilege 15 secret 1234
- R3(config)#line vty 0 4
- R3(config-line)#login local
- R3(config-line)#transport input ssh
- R3(config-line)#exit
- R2(config)#crypto key generate rsa (command)

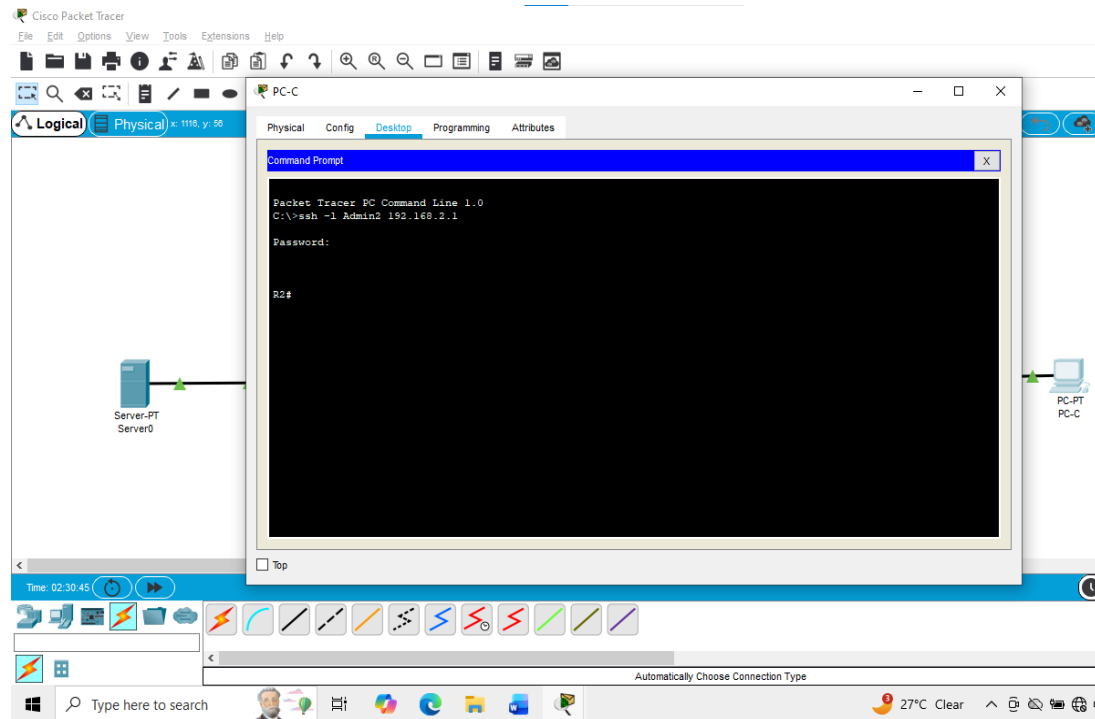


Step 10 : To check the connectivity of SSH

Go to PC-C -> Desktop -> Command prompt

And use command : `ssh -l Admin2 192.168.2.1`

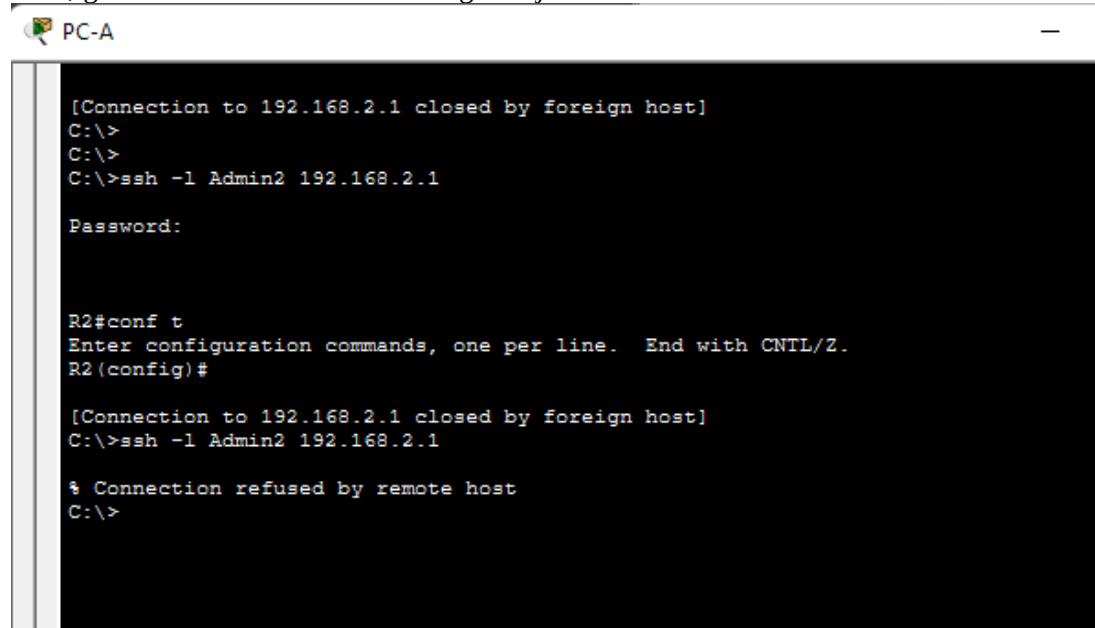
- Then it will ask for password
 - Enter the password which we have given during configuration of SSH (i.e in our case 1234)



If something like this shows up

Hence we can say that SSH is Configured Successfully.

Now, go to PC-A to check whether is gets rejected



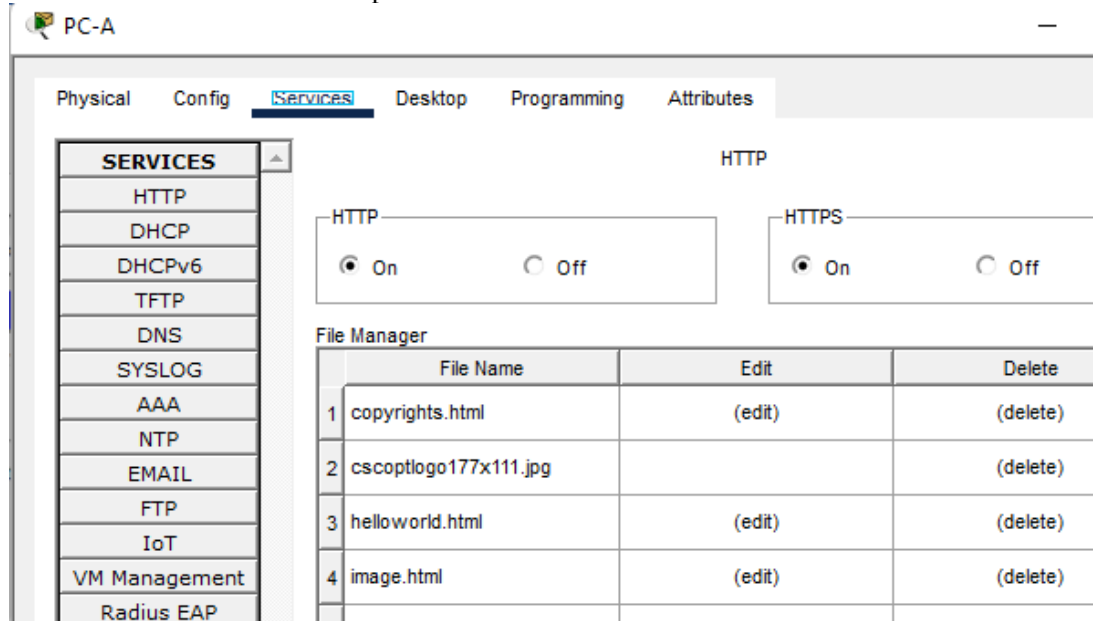
Here you can see that the PC-A has been refused access

4.C Configure ACLs on to mitigate attacks

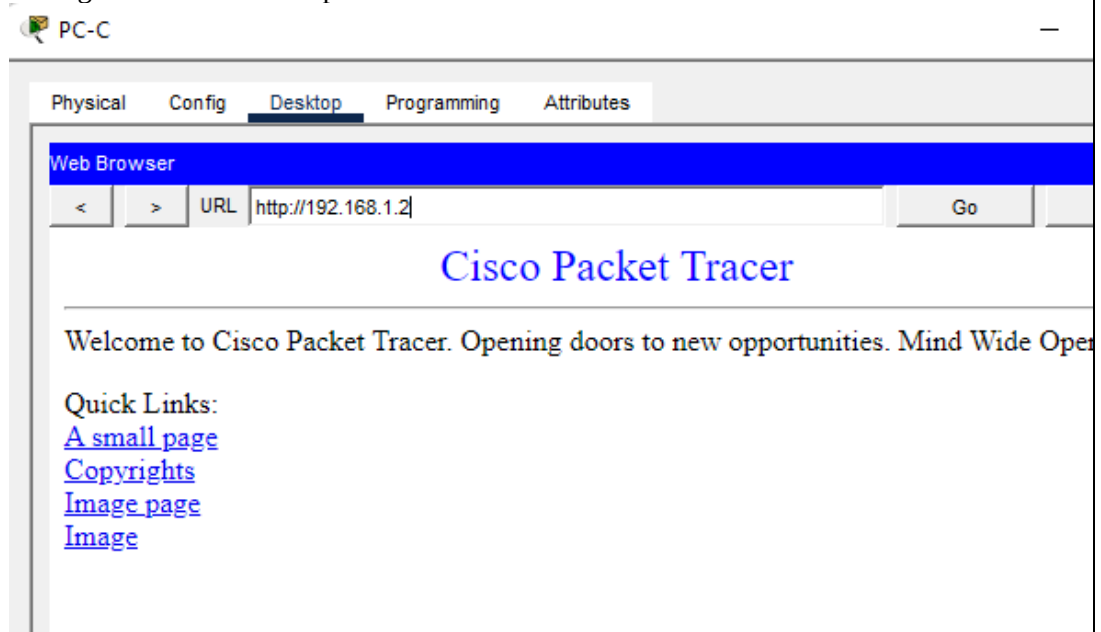
9-2-26

This is to filter out and deny access to devices not needed (i.e., to mitigate attacks)

Go to PC-A and turn on the http



Then go to PC-C > Desktop > web browser > and enter 192.168.1.2



Run these commands in R1 cli

```
R1>en
R1#config
Configuring from terminal, memory, or network [terminal]? t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 120 permit udp any host 192.168.1.2 eq domain
^
% Invalid input detected at '^' marker.

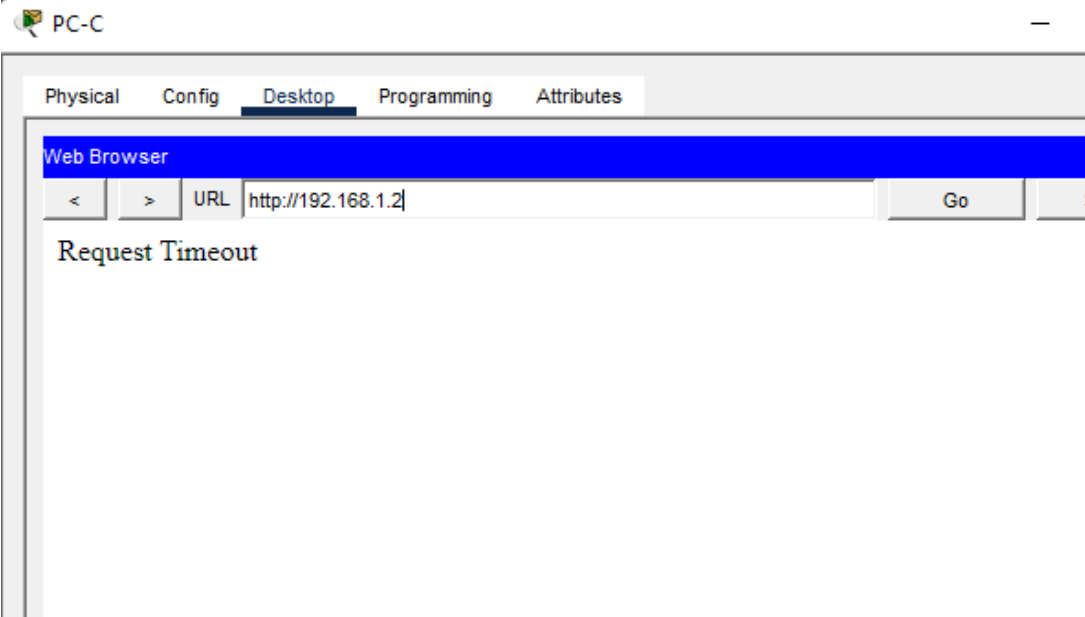
R1(config)#access-list 120 permit udp any host 192.168.1.2 eq domain
R1(config)#access-list 120 permit tcp any host 192.168.1.2 eq smtp
R1(config)#access-list 120 permit tcp any host 192.168.1.2 eq ftp
R1(config)#access-list 120 d tcp any host 192.168.1.2 eq ftp^Z
```

```

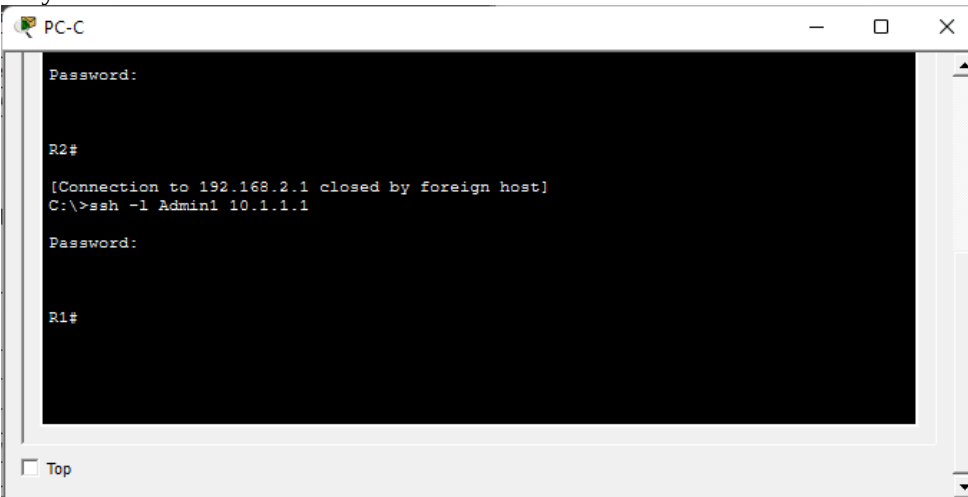
R1#
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 120 deny tcp any host 192.168.1.2 eq 443
R1(config)#access-list 120 permit tcp host 192.168.3.2 host 10.1.1.1 eq 22
R1(config)#interface s0/0/0
R1(config-if)#ip access-group 120 in
R1(config-if)#

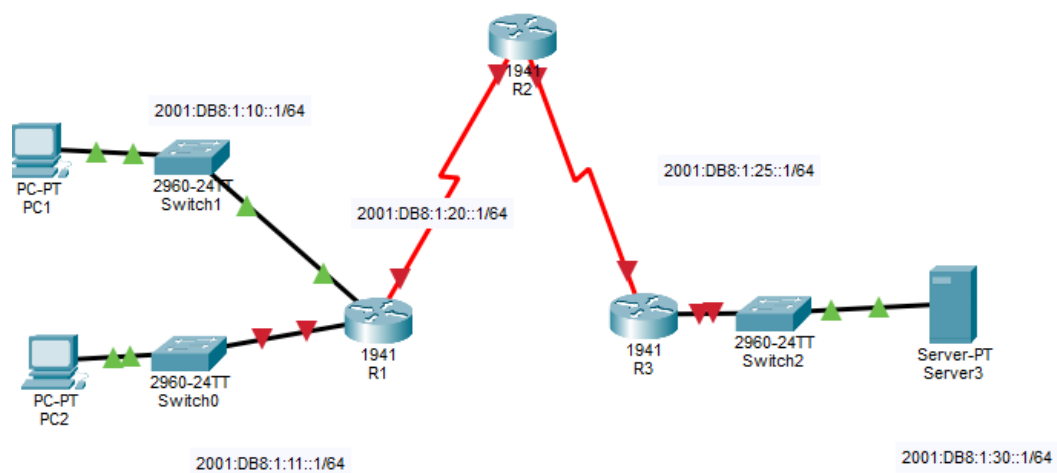
```

Then again go to PC-C and check if the access is denied



As you can see PC-C can still access R1





Addressing Table

Device	Interface	IPv6 Address/Prefix	Default Gateway
Server3	NIC	2001:DB8:1:30::30/64	FE80::30

```

R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ipv6 unicast-routing
R1(config)#int g0/0
R1(config-if)#ipv6 enable
R1(config-if)#no shut

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed s
up
R1(config-if)#ipv6 address 2001:DB8:1:10::1/64

```

PC1

Subnet Mask	
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	2001:DB8:1:10::2 /
Link Local Address	FE80::260:5CFF:FEA9:D1B3
Default Gateway	2001:DB8:1:10::1
DNS Server	



R1

%SYS-5-CONFIG_I: Configured from console by console

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ipv6 unicast-routing
R1(config)#int g0/1
R1(config-if)#ipv6 enable
R1(config-if)#no shut
```

```
R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

```
R1(config-if)#ipv6 address 2001:DB8:1:11::1/64
R1(config-if)#
```



PC2

Subnet Mask

Default Gateway

DNS Server

0.0.0.0

0.0.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

2001:DB8:1:11::2

Link Local Address

FE80::290:21FF:FE07:CA

Default Gateway

2001:DB8:1:11::1

DNS Server

802.1X



R3

R3#
%SYS-5-CONFIG_I: Configured from console by console

```
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ipv6 unicast-routing
R3(config)#int g0/0
R3(config-if)#ipv6 enable
R3(config-if)#no shut
```

```
R3(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

```
R3(config-if)#ipv6 address 2001:DB8:1:30::1/64
R3(config-if)#
```

Copy

Paste

☐ Top

☐ Automatic

☒ Static

IPv6 Address

2001:DB8:1:30::2

Link Local Address

FE80::2E0:F7FF:FE57:E7CB

Default Gateway

2001:DB8:1:30::1

DNS Server

802.1X

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int se0/1/0
R1(config-if)#ipv6 enable
R1(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
R1(config-if)#ipv6 address 2001:DB8:1:20::1/64
R1(config-if)#
```

```
R2>en
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ipv6 unicast-routing
R2(config)#int se0/1/0
R2(config-if)#ipv6 enable
R2(config-if)#ipv6 address 2001:DB8:1:20::1/64
R2(config-if)#
```

```
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ipv6 unicast-routing
R2(config)#int se0/1/1
R2(config-if)#ipv6 enable
R2(config-if)#no shut
R2(config-if)#ipv6 address 2001:DB8:1:25::1/64
R2(config-if)#
```

```
R3(config)#ipv6 unicast-routing
R3(config)#int se0/1/0
R3(config-if)#ipv6 enable
R3(config-if)#no shut

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
R3(config-if)#ipv6 address 2001:DB8:1:25::1/64
R3(config-if)#

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up
```

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ipv6 route 2001:DB8:1:25::/64 2001:DB8:1:20::2
R1(config)#ipv6 route 2001:DB8:1:30::/64 2001:DB8:1:20::2
```

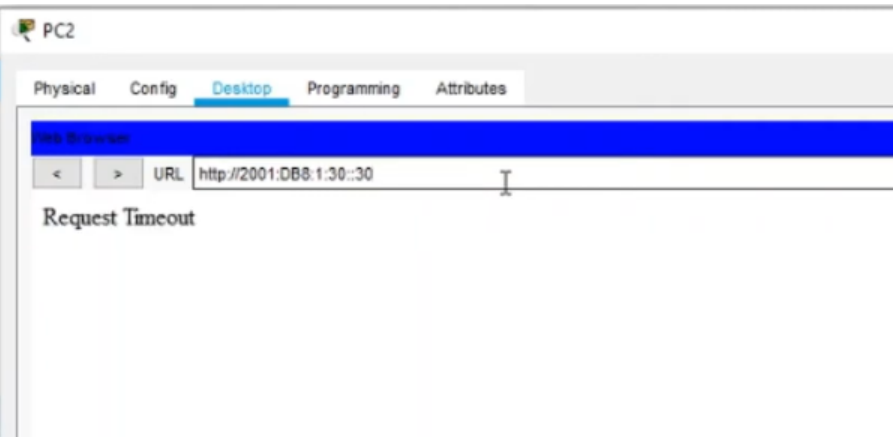
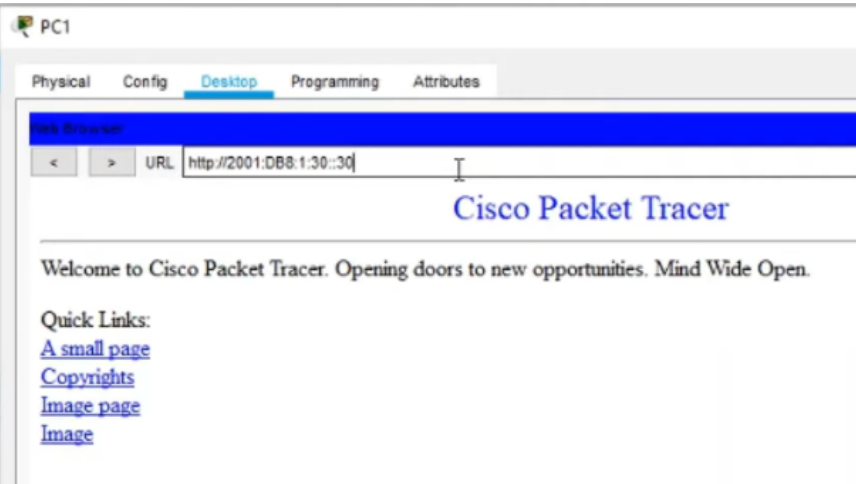
```
R2(config)#ipv6 route 2001:DB8:1:20::/64 2001:DB8:1:10::1
R2(config)#ipv6 route 2001:DB8:1:20::/64 2001:DB8:1:11::1
R2(config)#ipv6 route 2001:DB8:1:25::/64 2001:DB8:1:30::1
```

```
R3(config)#ipv6 route 2001:DB8:1:25::/64 2001:DB8:1:20::1
R3(config)#ipv6 route 2001:DB8:1:25::/64 2001:DB8:1:10::1
R3(config)#ipv6 route 2001:DB8:1:25::/64 2001:DB8:1:11::1
```

```

R1(config)#ipv6 access-list BLOCK_HTTP
R1(config-ipv6-acl)#deny tcp any host 2001:DB8:1:30::30
R1(config-ipv6-acl)#deny tcp any host 2001:DB8:1:30::30 eq www
R1(config-ipv6-acl)#deny tcp any host 2001:DB8:1:30::30 eq 443
R1(config-ipv6-acl)#permit ipv6 any any
R1(config-ipv6-acl)#exit
R1(config)#int g0/0
R1(config-if)#ipv6 traffic-filter BLOCK_HTTP
% Incomplete command.
R1(config-if)#ipv6 traffic-filter BLOCK_HTTP in
R1(config-if)#

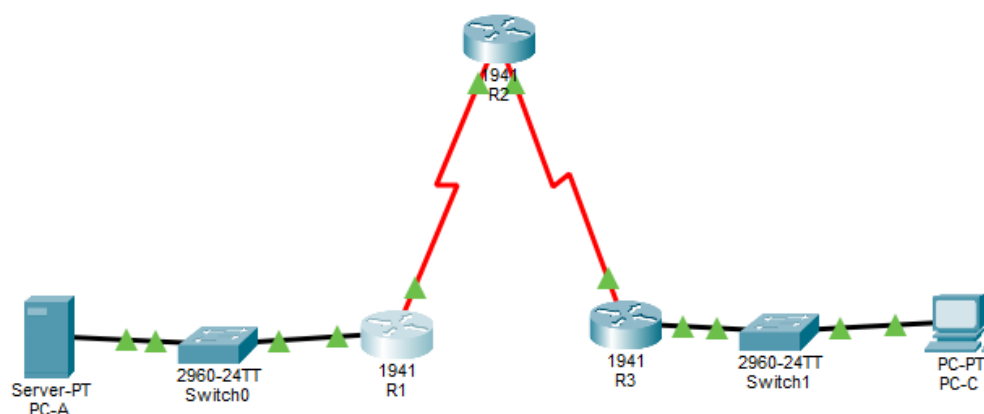
```



```

R3>en
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ipv6 access-list BLOCK_ICMP
R3(config-ipv6-acl)#deny icmp any any
R3(config-ipv6-acl)#permit ipv6 any any
R3(config-ipv6-acl)#exit
R3(config)#int g0/0
R3(config-if)#ipv6 traffic-filter BLOCK_ICMP out
R3(config-if)#

```



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A	N/A
R2	S0/0/0	10.1.1.2	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.1	255.255.255.252	N/A	N/A
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/6
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18

Objectives

- Verify connectivity among devices before firewall configuration.
- Configure a zone-based policy (ZPF) firewall on R3.
- Verify ZPF firewall functionality using ping, SSH, and a web browser.

```

R1(config)#line console 0
R1(config-line)#password conpa55
R1(config-line)#login
R1(config-line)#exit

R2(config)#line console 0
R2(config-line)#password conpa55
R2(config-line)#login
R2(config-line)#exit

R3(config)#line console 0
R3(config-line)#password conpa55
R3(config-line)#login
R3(config-line)#exit

R1(config)#line vty 0 4
R1(config-line)#password vtypa55
R1(config-line)#login
R1(config-line)#exit

R2(config)#line vty 0 4
R2(config-line)#password vtypa55
R2(config-line)#login
R2(config-line)#exit
  
```

```

R3(config)#line vty 0 4
R3(config-line)#password vtypass
R3(config-line)#login
R3(config-line)#exit

R1(config)#enable secret enpass
R1(config)#exit
R2(config)#enable secret enpass
R2(config)#exit
R3(config)#enable secret enpass
R3(config)#exit

R1(config)#ip domain-name ccnasecurity.com
R1(config)#username admin secret adminpass
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#exit
R2(config)#ip domain-name ccnasecurity.com
R2(config)#username admin secret adminpass
R2(config)#line vty 0 4
R2(config-line)#login local
R2(config-line)#exit
R3(config)#ip domain-name ccnasecurity.com
R3(config)#username admin secret adminpass
R3(config)#line vty 0 4
R3(config-line)#login local
R3(config-line)#exit

R1(config)#crypto key generate rsa
The name for the keys will be: R1.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R2(config)#crypto key generate rsa
The name for the keys will be: R2.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R3(config)#crypto key generate rsa
The name for the keys will be: R3.ccnasecurity.com
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#ip route 10.2.2.0 255.255.255.252 10.1.1.2
*Mar 1 0:17:49.764: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#ip route 192.168.3.0 255.255.255.0 10.1.1.2
R2(config)#ip route 192.168.1.0 255.255.255.0 10.1.1.1
*Mar 1 0:18:12.51: %SSH-5-ENABLED: SSH 1.99 has been enabled
R2(config)#ip route 192.168.3.0 255.255.255.0 10.2.2.1
R3(config)#ip route 192.168.1.0 255.255.255.0 10.2.2.2
*Mar 1 0:18:30.769: %SSH-5-ENABLED: SSH 1.99 has been enabled
R3(config)#ip route 10.2.2.0 255.255.255.252 10.2.2.2

```

```

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#license boot module c1900 technology-package securityk9
R3#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
R3#reload

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#zone security IN-ZONE
R3(config-sec-zone)#exit
R3(config)#
R3(config)#zone security OUT-ZONE
R3(config-sec-zone)#exit
R3(config)#
R3(config)#access-list 101 permit ip 192.168.3.0 0.0.0.255 any
R3(config)#class-map type inspect match-all IN-NET-CLASS-MAP
R3(config-cmap)#match access-group 101
R3(config-cmap)#exit
R3(config)#
R3(config)#policy-map type inspect IN-2-OUT-PMAP
R3(config-pmap)#class type inspect IN-NET-CLASS-MAP
R3(config-pmap-c)#inspect
%No specific protocol configured in class IN-NET-CLASS-MAP for
inspection. All protocols will be inspected
R3(config-pmap-c)#exit
R3(config-pmap)#exit
R3(config)#zone-pair security IN-2-OUT-ZPAIR source IN-ZONE
destination OUT-ZONE
R3(config-sec-zone-pair)#service-policy type inspect IN-2-OUT-PAMP
Policy IN-2-OUT-PAMP does not exist
Inspect service-policy attachment failed
R3(config-sec-zone-pair)#service-policy type inspect IN-2-OUT-PMAP
R3(config-sec-zone-pair)#exit
R3(config)#
R3(config)#int gig0/0
R3(config-if)#zone-member security IN-ZONE
R3(config-if)#EXIT
R3(config)#int se0/1/0
R3(config-if)#zone-member security OUT-ZONE
R3(config-if)#exit
R3#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
R3#

```

```
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

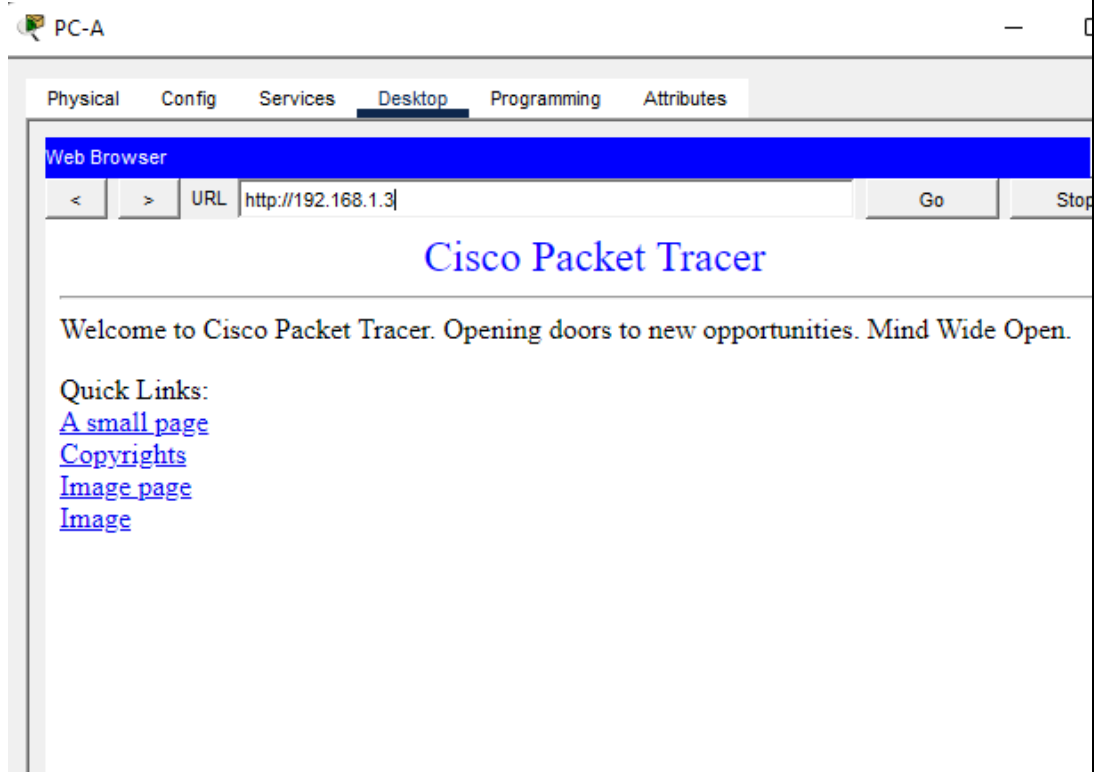
Reply from 192.168.1.3: bytes=32 time=10ms TTL=125
Reply from 192.168.1.3: bytes=32 time=2ms TTL=125
Reply from 192.168.1.3: bytes=32 time=2ms TTL=125
Reply from 192.168.1.3: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 10ms, Average = 4ms

C:\>ssh -l admin 10.2.2.2

Password:

R2>
```



```
R3#show policy-map type inspect zone-pair sessions
[Connection to 10.2.2.2 closed by foreign host]
C:\>ssh -l admin 10.2.2.2

Password:

R2>
```

PC-A

```
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

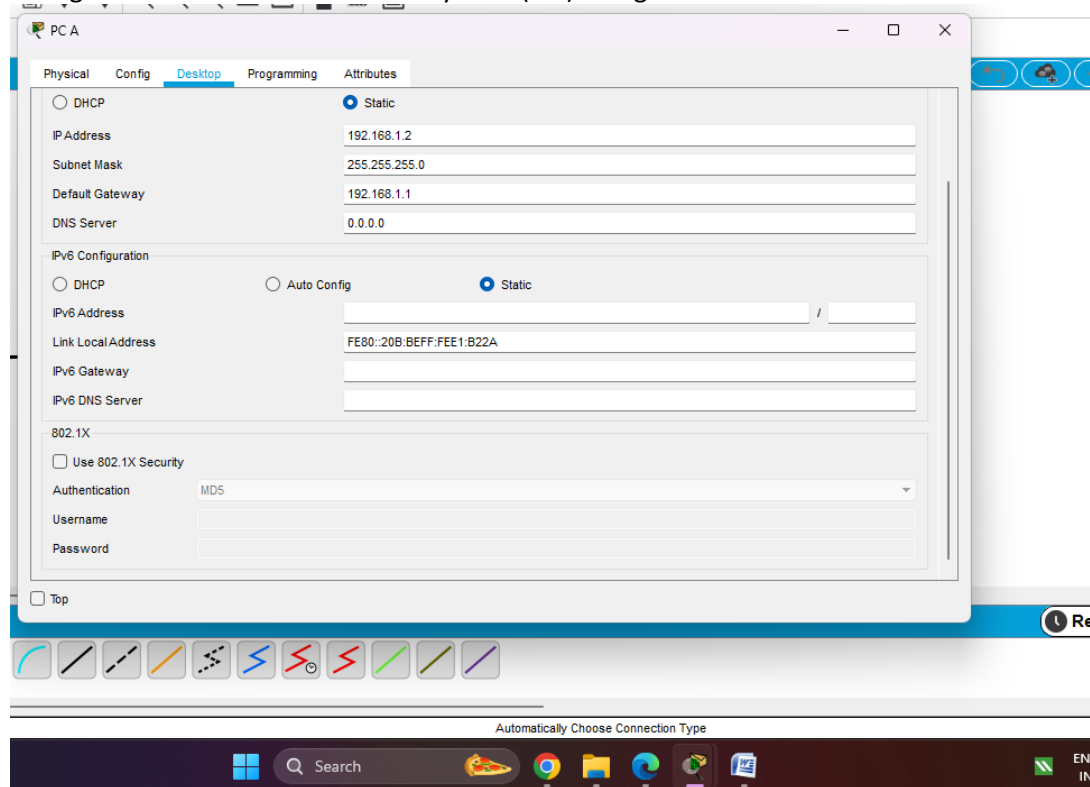
Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
R2#ping 192.168.3.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.3, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
```

Practical
6

Configure IOS Intrusion Prevention System (IPS) Using the CLI

10-2-26



Syslog

Physical

Config

Services

Desktop

Programming

Attributes

DHCP

Static

IP Address

192.168.1.50

Subnet Mask

255.255.255.0

Default Gateway

192.168.1.1

DNS Server

0.0.0.0

IPv6 Configuration

DHCP

Auto Config

Static

IPv6 Address

/

Link Local Address

FE80::260:3EFF:FE3B:7B59

IPv6 Gateway

IPv6 DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

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R1

Physical

Config

CLI

Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

Serial0/1/0

Serial0/1/1

GigabitEthernet0/1

Port Status ☒ On

Bandwidth ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0001.C7D3.1102

IP Configuration

IP Address 192.168.1.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Equivalent IOS Commands

Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/1
Router(config-if)#

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R1

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

Serial0/1/0

Serial0/1/1

Serial0/0/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 2000000

IP Configuration

IP Address 10.1.1.1

Subnet Mask 255.255.255.252

Tx Ring Limit 10

Equivalent IOS Commands

```
Router(config)#interface Serial0/0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#
```

R1

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Serial0/0/0

Serial0/0/1

Serial0/1/0

Serial0/1/1

Serial0/0/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 2000000

IP Configuration

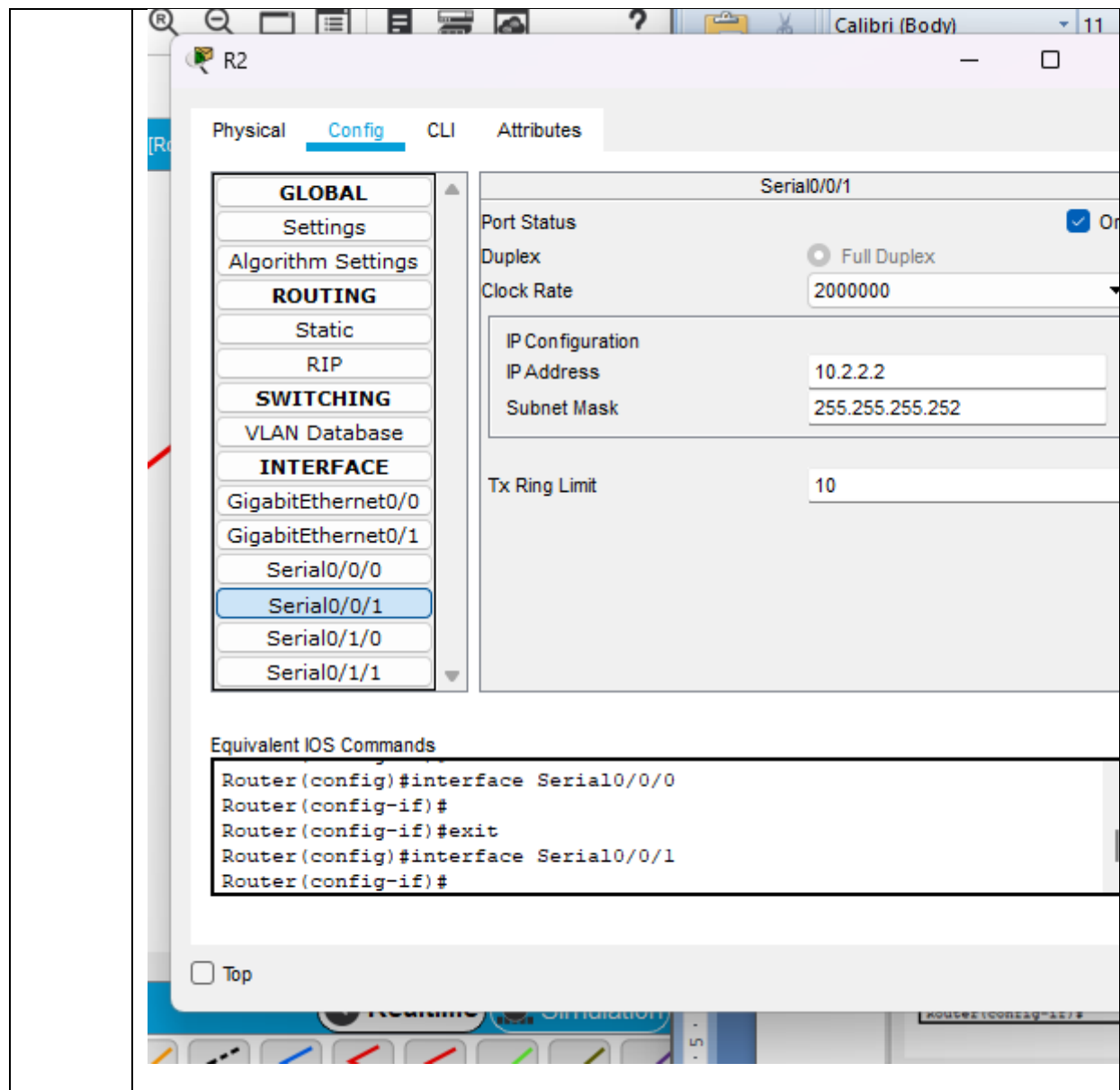
IP Address 10.1.1.1

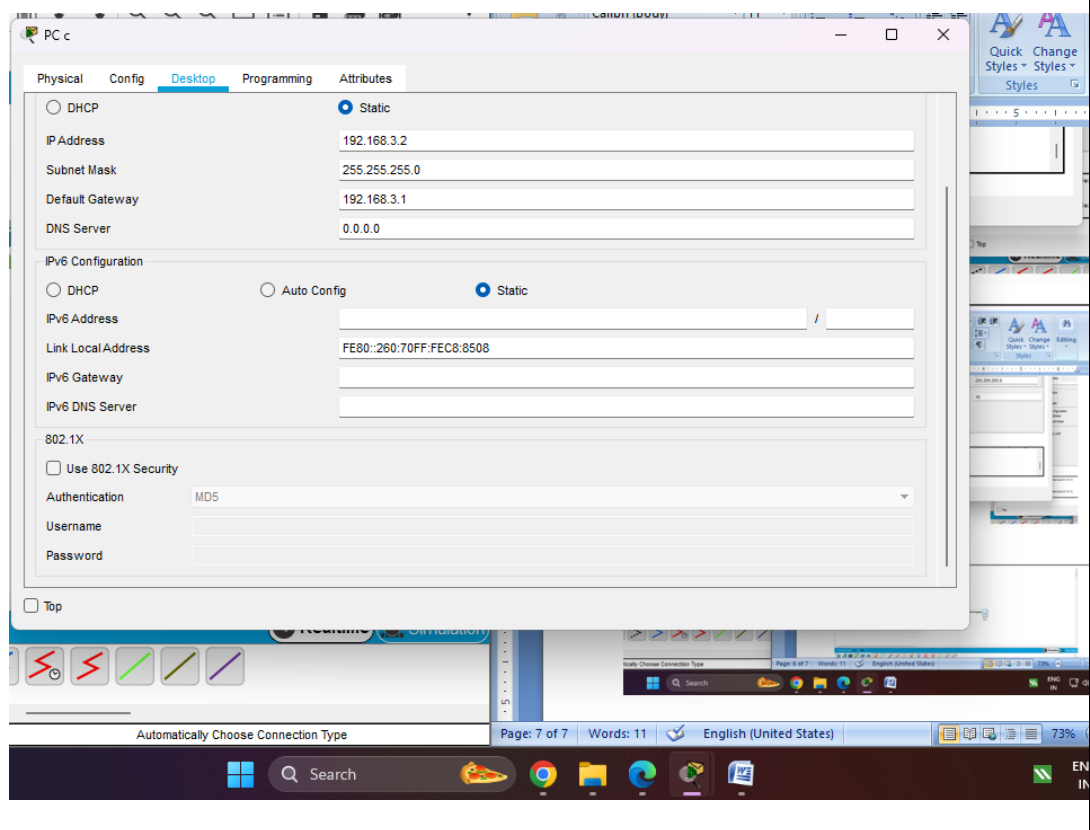
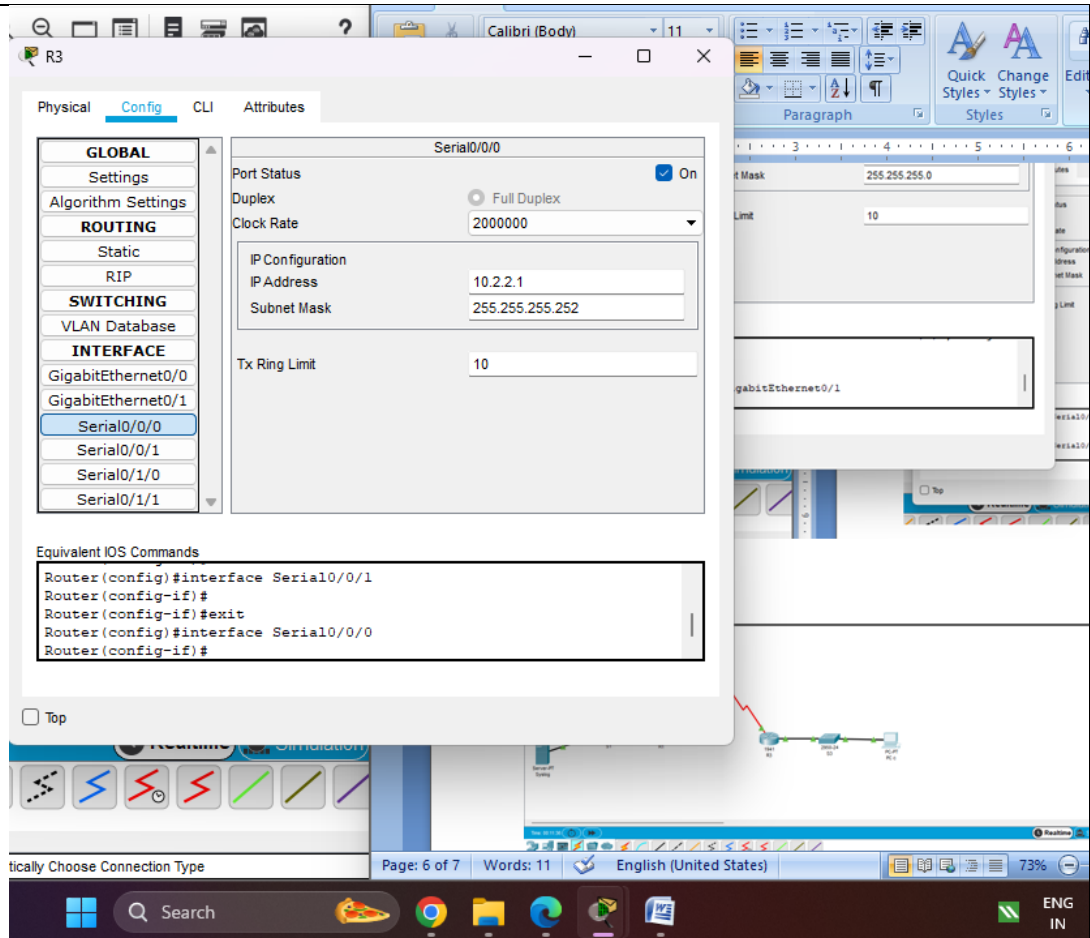
Subnet Mask 255.255.255.252

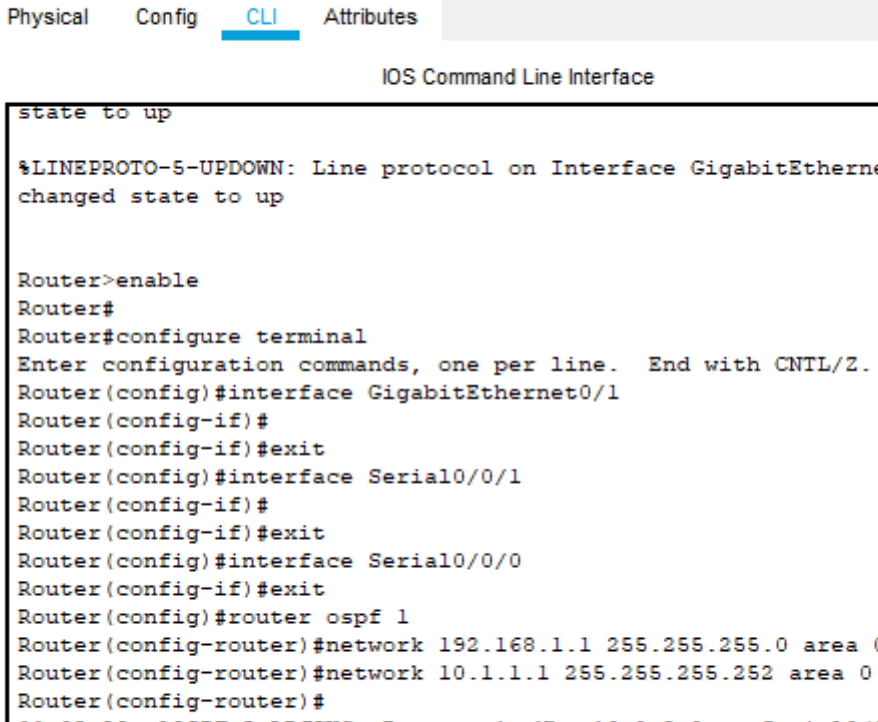
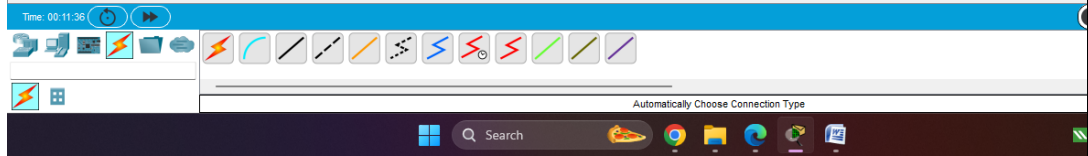
Tx Ring Limit 10

Equivalent IOS Commands

```
Router(config)#interface Serial0/0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#
```







```
state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to up

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.1 255.255.255.0 area 0
Router(config-router)#network 10.1.1.1 255.255.255.252 area 0
Router(config-router)#
00:05:29: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/0 from
LOADING to FULL, Loading Done
```

R2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/1
Router(config-if)#exit
Router(config)#router ospf 1
Router(config-router)#network 10.1.1.2 255.255.255.252 area 0
Router(config-router)#network 10.2.2.2 2
00:08:18: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.1.1 on Serial0/0/1
from LOADING to Fnetwork 10.2.2.2 2
^
% Invalid input detected at '^' marker.
Router(config-router)#network 10.2.2.2 2 255.255.255.252 area 0
^
% Invalid input detected at '^' marker.
Router(config-router)#network 10.2.2.2 255.255.255.252 area 0
Router(config-router)#
00:08:57: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.2.1 on Serial0/0/1
from LOADING to Fnetwork 10.2.2.2 255.255.255.252 area 0
```

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R3

Physical Config **CLI** Attributes

IOS Command Line Interface

```

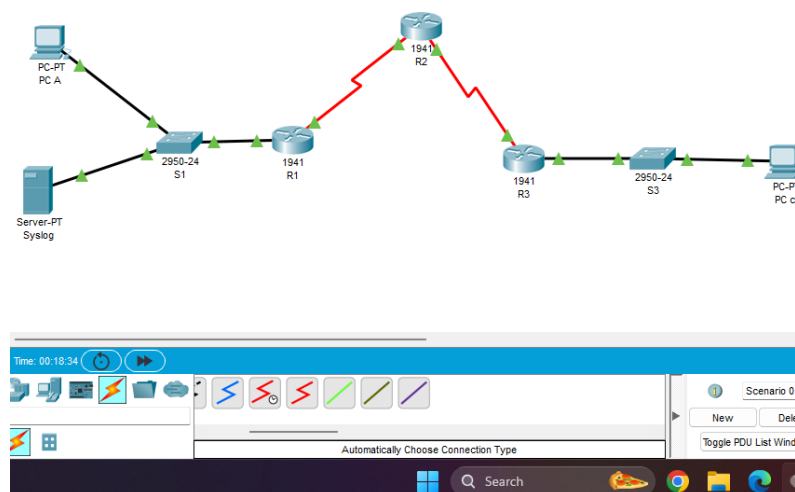
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed
state to up

Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#exit
Router(config)#network 192.168.3.1 255.255.255.0 area 0
^
% Invalid input detected at '^' marker.

Router(config)#router ospf
% Incomplete command.
Router(config)#router ospf 1
Router(config-router)#network 192.168.3.1 255.255.255.0 area 0
Router(config-router)#network 10.2.2.1 255.255.255.0 area 0
Router(config-router)#
00:08:48: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/0 from

```

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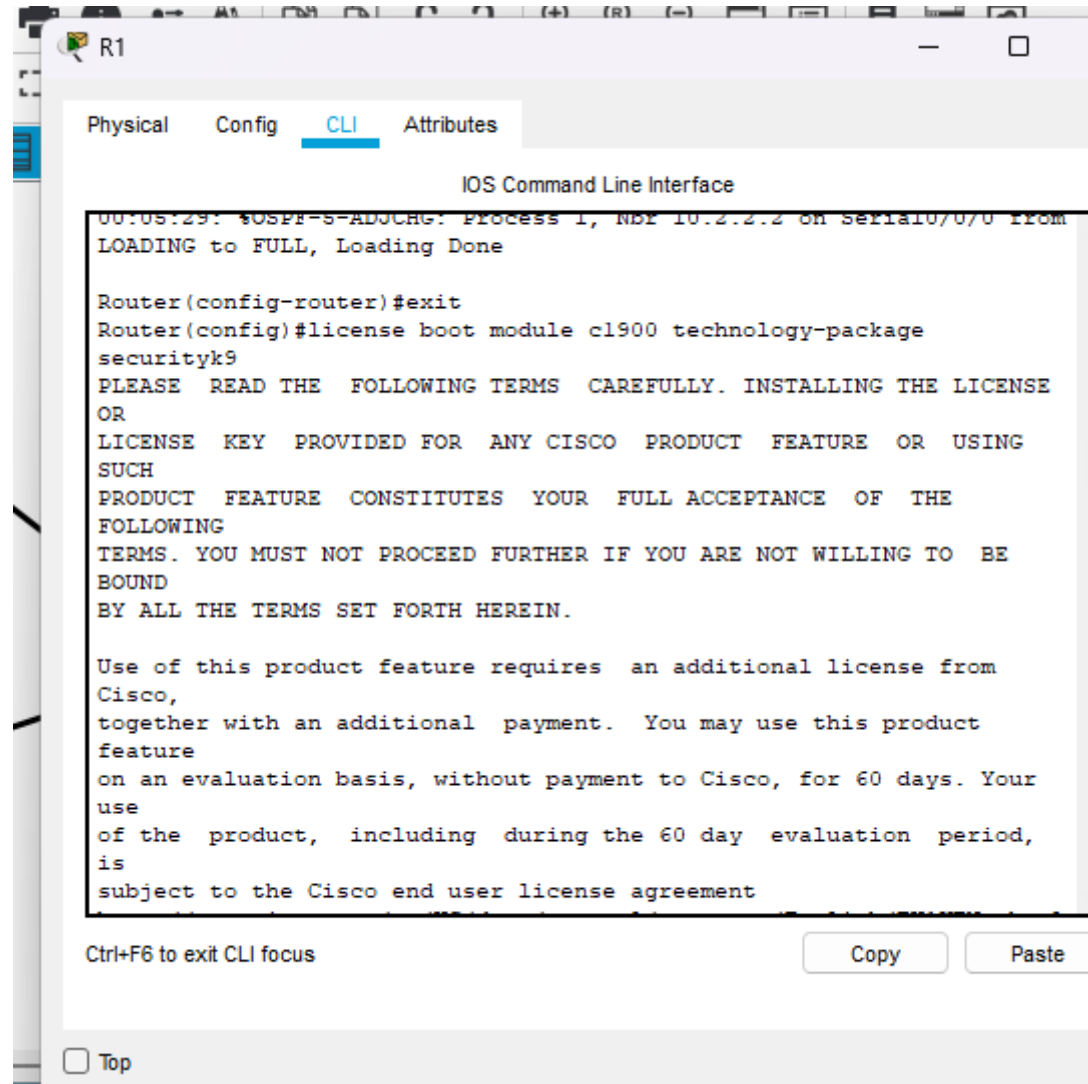


Part 1: Enable IOS IPS

Note: Within Packet Tracer, the routers already have the signature files imported and in place. They are the default xml files in flash. For this reason, it is not necessary to configure the public crypto key and complete a manual import of the signature files.

Step 1: Enable the Security Technology package.

- a. On R1, issue the show version command to view the Technology Package license information.
- b. If the Security Technology package has not been enabled, use the following command to enable the package.
R1(config)# license boot module c1900 technology-package securityk9
- c. Accept the end user license agreement.
- d. Save the running-config and reload the router to enable the security license.
- e. Verify that the Security Technology package has been enabled by using the show version command.



```
00:05:29: %OSPF-5-ADJCHG: Process 1, Nbr 10.2.2.2 on Serial0/0/0 from
LOADING to FULL, Loading Done

Router(config-router)#exit
Router(config)#license boot module c1900 technology-package
securityk9
PLEASE READ THE FOLLOWING TERMS CAREFULLY. INSTALLING THE LICENSE
OR
LICENSE KEY PROVIDED FOR ANY CISCO PRODUCT FEATURE OR USING
SUCH
PRODUCT FEATURE CONSTITUTES YOUR FULL ACCEPTANCE OF THE
FOLLOWING
TERMS. YOU MUST NOT PROCEED FURTHER IF YOU ARE NOT WILLING TO BE
BOUND
BY ALL THE TERMS SET FORTH HEREIN.

Use of this product feature requires an additional license from
Cisco,
together with an additional payment. You may use this product
feature
on an evaluation basis, without payment to Cisco, for 60 days. Your
use
of the product, including during the 60 day evaluation period,
is
subject to the Cisco end user license agreement
```

Ctrl+F6 to exit CLI focus

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☐ Top

```
Router#write
Building configuration...
[OK]
Router#
Router#reload
Proceed with reload? [confirm]
System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO1941/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 64/-1(On-board/DIMM0) bit mode with ECC
disabled
```

Step 2: Verify network connectivity.

a. Ping from PC-C to PC-A. The ping should be successful.

```
Command Prompt

Pingging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time=3ms TTL=125
Reply from 192.168.1.2: bytes=32 time=4ms TTL=125
Reply from 192.168.1.2: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 3ms

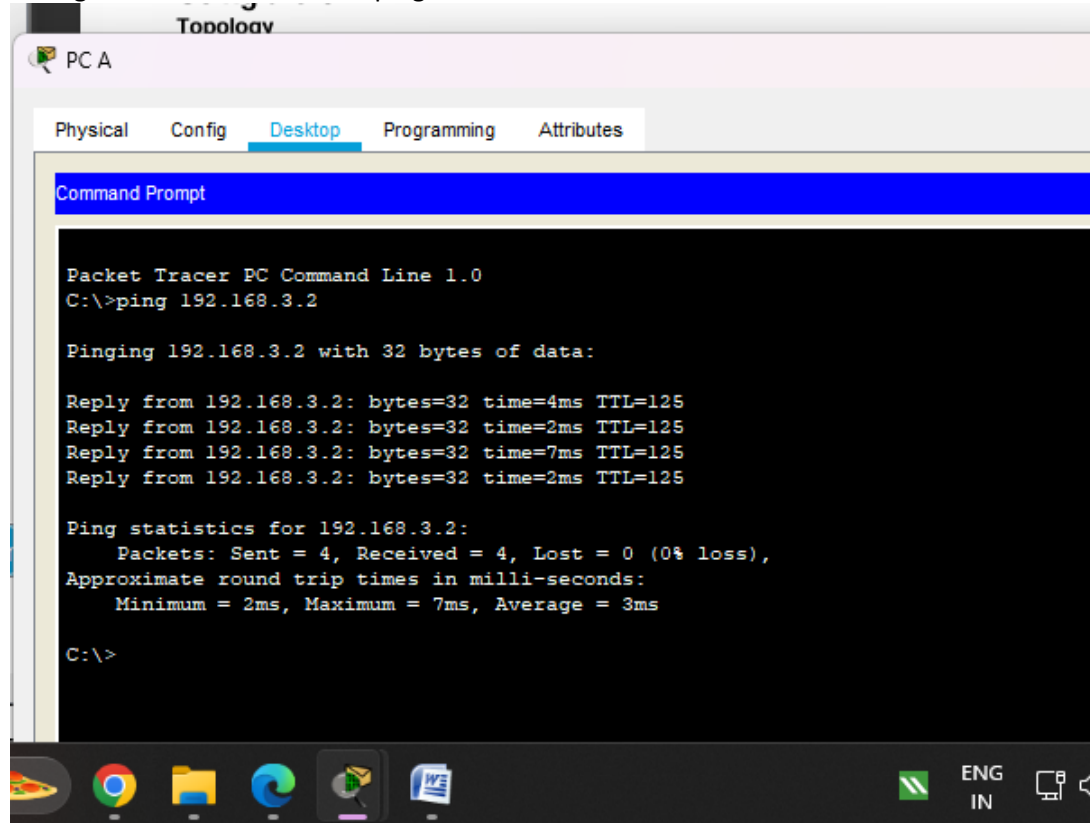
C:\>ping 192.168.1.2

Pingging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=7ms TTL=125
Reply from 192.168.1.2: bytes=32 time=2ms TTL=125
Reply from 192.168.1.2: bytes=32 time=5ms TTL=125
Reply from 192.168.1.2: bytes=32 time=5ms TTL=125

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 7ms, Average = 4ms
```

b. Ping from PC-A to PC-C. The ping should be successful.



Step 3: Create an IOS IPS configuration directory in flash. On R1, create a directory in flash using the mkdir command. Name the directory ipsdir.

R1# mkdir ipsdir

Create directory filename [ipsdir]? <Enter> Created
dir flash:ipsdir

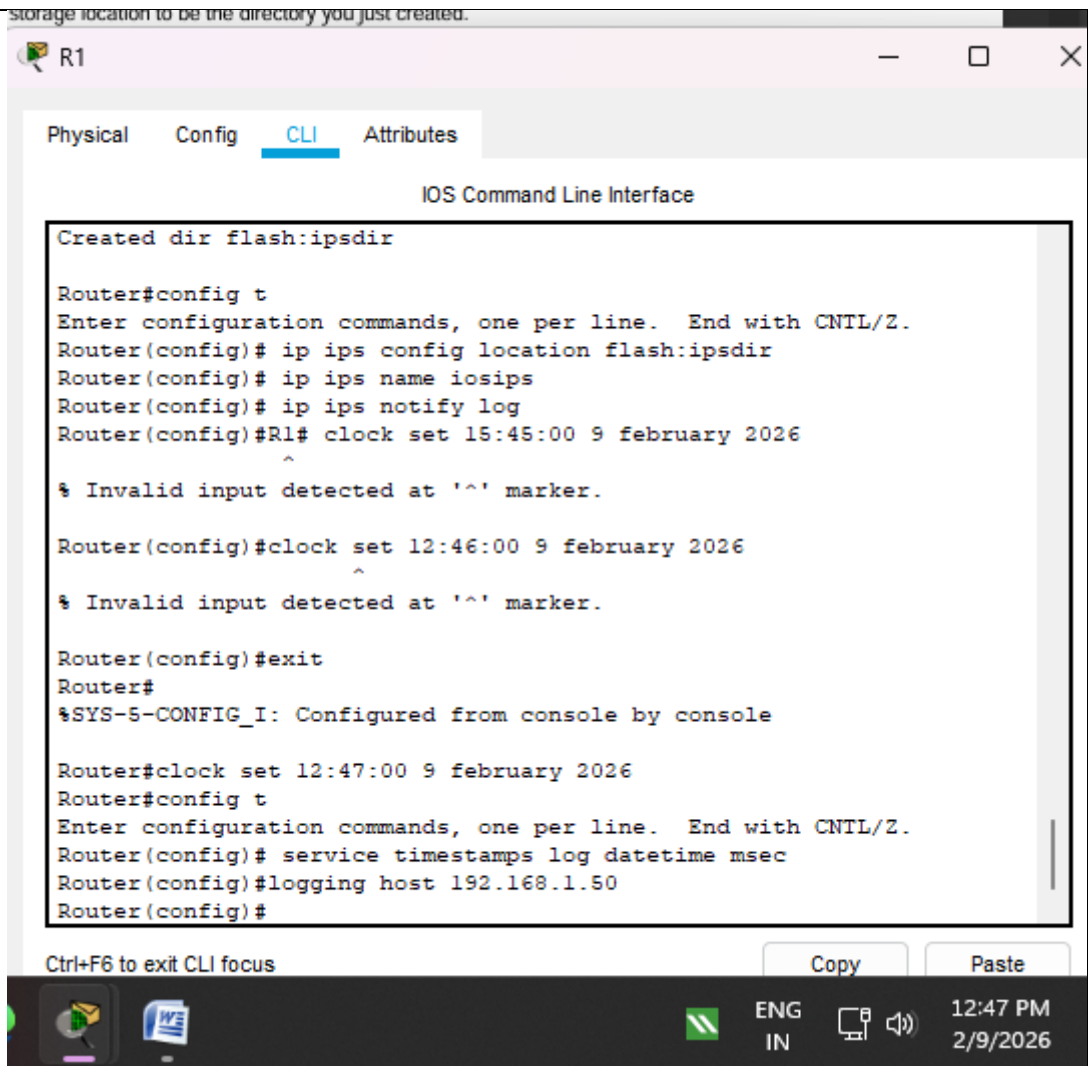
```
create directory filename ipsdir/? <Enter> Created
R1
Physical Config CLI Attributes
IOS Command Line Interface
data          disable      None          None
Configuration register is 0x2102
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#mkdir ipsdir
Create directory filename [ipsdir]?Created
Created dir flash:Created
Router#mkdir ipsdir
Create directory filename [ipsdir]?
Created dir flash:ipsdir
```

Step 4: Configure the IPS signature storage location. On R1, configure the IPS signature storage location to be the directory you just created.
R1(config)# ip ips config location flash:ipsdir

Step 5: Create an IPS rule.
On R1, create an IPS rule name using the ip ips name name command in global configuration mode. Name the IPS rule iosips.
R1(config)# ip ips name iosips

Step 6: Enable logging.
IOS IPS supports the use of syslog to send event notification. Syslog notification is enabled by default. If logging console is enabled, IPS syslog messages display. a. Enable syslog if it is not enabled.
R1(config)# ip ips notify log

storage location to be the directory you just created.



```
Created dir flash:ipsdir

Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)# ip ips config location flash:ipsdir
Router(config)# ip ips name iosips
Router(config)# ip ips notify log
Router(config)#R1# clock set 15:45:00 9 february 2026
^
% Invalid input detected at '^' marker.

Router(config)#clock set 12:46:00 9 february 2026
^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#clock set 12:47:00 9 february 2026
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)# service timestamps log datetime msec
Router(config)#logging host 192.168.1.50
Router(config)#
```

Ctrl+F6 to exit CLI focus

Copy Paste

ENG IN 12:47 PM 2/9/2026

Step 7: Configure IOS IPS to use the signature categories.

Retire the all signature category with the retired true command (all signatures within the signature release).

Unretire the IOS_IPS Basic category with the retired false command. R1(config)# ip ips signature category

R1(config-ips-category)# category all

R1(config-ips-category-action)# retired true

R1(config-ips-category-action)# exit

R1(config-ips-category)# category ios_ips basic

R1(config-ips-category-action)# retired false

R1(config-ips-category-action)# exit

R1(config-ips-cateogry)# exit

Do you want to accept these changes? [confirm] <Enter>

```
Router#clock set 12:47:00 9 february 2026
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)# service timestamps log datetime msec
Router(config)# logging host 192.168.1.50
Router(config)# ip ips signature
% Ambiguous command: "ip ips signature"
Router(config)# ip ips signature
% Ambiguous command: "ip ips signature"
Router(config)# ip ips signature-category
Router(config-ips-category)# category all
Router(config-ips-category-action)# retired true
Router(config-ips-category-action)# exit
Router(config-ips-category)# category ios_ips basic
Router(config-ips-category-action)# retired false
Router(config-ips-category-action)# exit
Router(config-ips-category)# exit
Do you want to accept these changes? [confirm]
Applying Category configuration to signatures ...
%IPS-6-ENGINE_BUILDING: atomic-ip - 288 signatures - 6 of 13 engines
%IPS-6-ENGINE_READY: atomic-ip - build time 30 ms - packets for this
engine will be scanned
```

Step 8: Apply the IPS rule to an interface.

Apply the IPS rule to an interface with the `ip ips name direction` command in interface configuration mode.

Apply the rule outbound on the G0/1 interface of R1. After you enable IPS, some log messages will be sent to the console line indicating that the IPS engines are being initialized.

Note: The direction in means that IPS inspects only traffic going into the interface.

Similarly, out means that

IPS inspects only traffic going out of the interface.

R1(config)# interface g0/1

R1(config-if)# ip ips iosips out

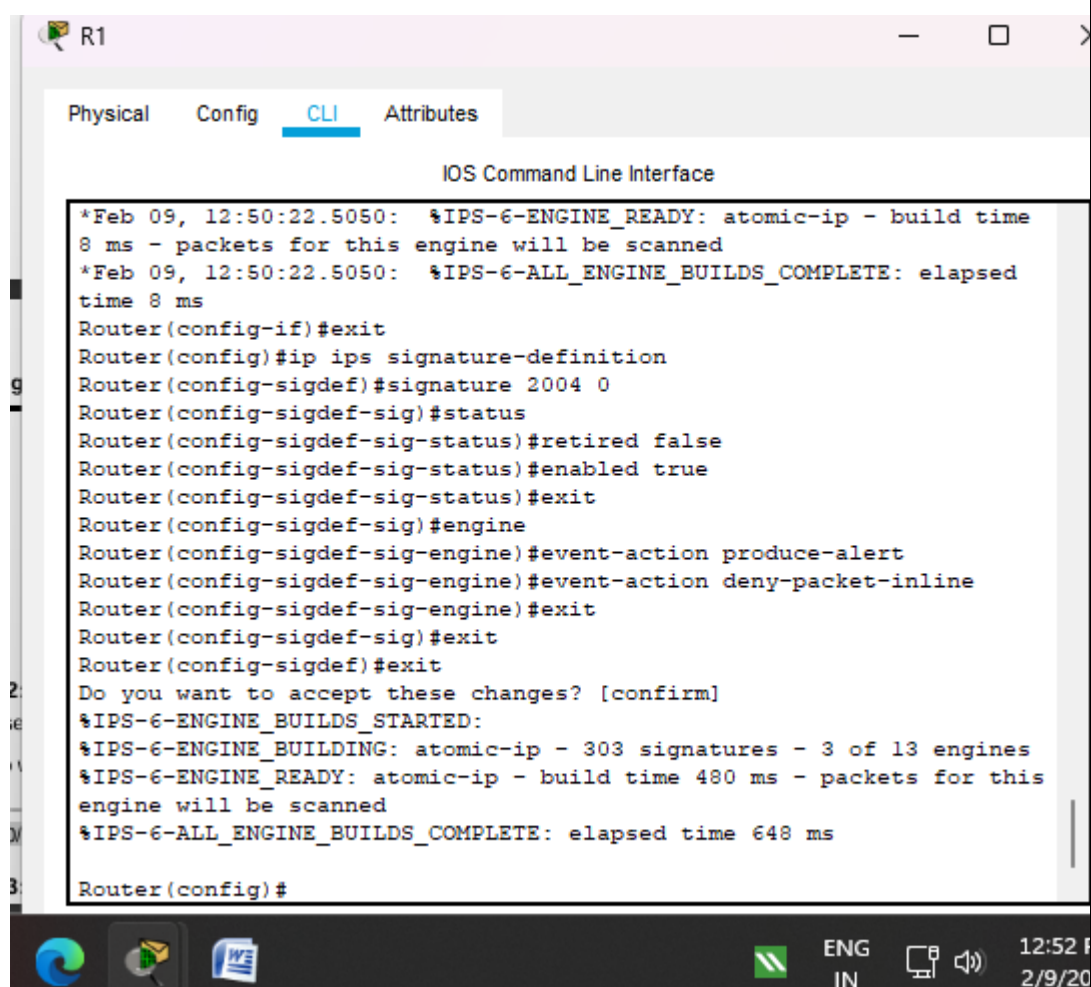
```
Router(config)# interface g0/1
Router(config-if)# ip ips iosips out
Router(config-if)#
*Feb 09, 12:50:22.5050: %IPS-6-ENGINE_BUILDS_STARTED: 12:50:22 UTC
Feb 09 2026
*Feb 09, 12:50:22.5050: %IPS-6-ENGINE_BUILDING: atomic-ip - 3
signatures - 1 of 13 engines
*Feb 09, 12:50:22.5050: %IPS-6-ENGINE_READY: atomic-ip - build time
8 ms - packets for this engine will be scanned
*Feb 09, 12:50:22.5050: %IPS-6-ALL_ENGINE_BUILDS_COMPLETE: elapsed
time 8 ms
```

6B : Modify the Signature

10-2-26

Step 1: Change the event-action of a signature.

Un-retire the echo request signature (signature 2004, subsig ID 0), enable it, and change the signature action to alert and drop.



```
IOS Command Line Interface

*Feb 09, 12:50:22.5050: %IPS-6-ENGINE_READY: atomic-ip - build time
8 ms - packets for this engine will be scanned
*Feb 09, 12:50:22.5050: %IPS-6-ALL_ENGINE_BUILDS_COMPLETE: elapsed
time 8 ms
Router(config-if)#exit
Router(config)#ip ips signature-definition
Router(config-sigdef)#signature 2004 0
Router(config-sigdef-sig)#status
Router(config-sigdef-sig-status)#retired false
Router(config-sigdef-sig-status)#enabled true
Router(config-sigdef-sig-status)#exit
Router(config-sigdef-sig)#engine
Router(config-sigdef-sig-engine)#event-action produce-alert
Router(config-sigdef-sig-engine)#event-action deny-packet-inline
Router(config-sigdef-sig-engine)#exit
Router(config-sigdef-sig)#exit
Router(config-sigdef)#exit
Do you want to accept these changes? [confirm]
%IPS-6-ENGINE_BUILDS_STARTED:
%IPS-6-ENGINE_BUILDING: atomic-ip - 303 signatures - 3 of 13 engines
%IPS-6-ENGINE_READY: atomic-ip - build time 480 ms - packets for this
engine will be scanned
%IPS-6-ALL_ENGINE_BUILDS_COMPLETE: elapsed time 648 ms

Router(config)#
```

Step 2: Use show commands to verify IPS.

Use the show ip ips all command to view the IPS configuration status summary.

R1

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
Router#
*Feb 09, 12:53:23.5353: SYS-5-CONFIG_I: Configured from console by
console
*Feb 09, 12:53:23.5353: %SYS-6-LOGGINGHOST_STARTSTOP: Logging to
host 192.168.1.50 port 514 started - CLI initiated
Router#show ip ips all
IPS Signature File Configuration Status
  Configured Config Locations: flash:ipsdir
  Last signature default load time:
  Last signature delta load time:
  Last event action (SEAP) load time: -none-

  General SEAP Config:
  Global Deny Timeout: 3600 seconds
  Global Overrides Status: Enabled
  Global Filters Status: Enabled

IPS Auto Update is not currently configured

IPS Syslog and SDEE Notification Status
  Event notification through syslog is enabled
  Event notification through SDEE is enabled

IPS Signature Status
  Total Active Signatures: 1
```

Ctrl+F6 to exit CLI focus

CopyPaste

☐ Top

Step 3: Verify that IPS is working properly.

a. From PC-C, attempt to ping PC-A. Were the pings successful? Explain.

PC C

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 7ms, Average = 4ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Taskbar: File Explorer, Chrome, Edge, OneDrive, Word, ENG IN

It should fail.

b. From PC-A, attempt to ping PC-C. Were the pings successful? Explain.

PC A

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time=4ms TTL=125
Reply from 192.168.3.2: bytes=32 time=2ms TTL=125
Reply from 192.168.3.2: bytes=32 time=7ms TTL=125
Reply from 192.168.3.2: bytes=32 time=2ms TTL=125

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 7ms, Average = 3ms

C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time=5ms TTL=125
Reply from 192.168.3.2: bytes=32 time=6ms TTL=125
Reply from 192.168.3.2: bytes=32 time=2ms TTL=125
Reply from 192.168.3.2: bytes=32 time=3ms TTL=125

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 6ms, Average = 4ms

C:\>
```

Taskbar: Search, File Explorer, Chrome, Edge, OneDrive, Word, ENG IN

The ping should be successful. This is because the IPS rule does not cover echo reply.
When PC-A pings
PC-C, PC-C responds with an echo reply.

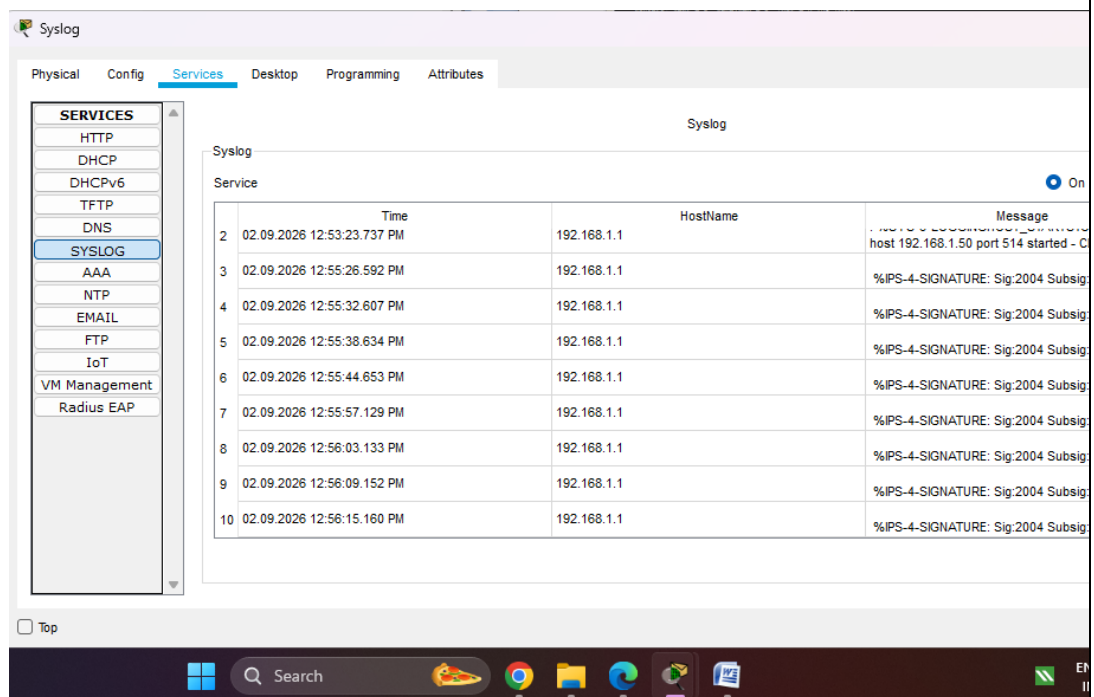
Step 4: View the syslog messages.

- Click the Syslog server.
- Select the Services tab.
- In the left navigation menu, select SYSLOG to view the log file.

Step 5: Check results.

Your completion percentage should be 100%. Click Check Results to see feedback and verification of which required components have been completed.

Output:



The screenshot shows the Syslog configuration window with the 'Services' tab selected. In the left-hand 'SERVICES' list, 'SYSLOG' is highlighted. The main area displays the 'Syslog' configuration, which is turned 'On'. Below this, a table lists 10 log entries. The table has three columns: 'Time', 'HostName', and 'Message'. The first entry (index 2) shows a message about host 192.168.1.50 port 514 starting. The subsequent 9 entries (indices 3-10) all show the same message: '%IPS-4-SIGNATURE: Sig:2004 Subsig:'.

	Time	HostName	Message
2	02.09.2026 12:53:23.737 PM	192.168.1.1	host 192.168.1.50 port 514 started - C
3	02.09.2026 12:55:26.592 PM	192.168.1.1	%IPS-4-SIGNATURE: Sig:2004 Subsig:
4	02.09.2026 12:55:32.607 PM	192.168.1.1	%IPS-4-SIGNATURE: Sig:2004 Subsig:
5	02.09.2026 12:55:38.634 PM	192.168.1.1	%IPS-4-SIGNATURE: Sig:2004 Subsig:
6	02.09.2026 12:55:44.653 PM	192.168.1.1	%IPS-4-SIGNATURE: Sig:2004 Subsig:
7	02.09.2026 12:55:57.129 PM	192.168.1.1	%IPS-4-SIGNATURE: Sig:2004 Subsig:
8	02.09.2026 12:56:03.133 PM	192.168.1.1	%IPS-4-SIGNATURE: Sig:2004 Subsig:
9	02.09.2026 12:56:09.152 PM	192.168.1.1	%IPS-4-SIGNATURE: Sig:2004 Subsig:
10	02.09.2026 12:56:15.160 PM	192.168.1.1	%IPS-4-SIGNATURE: Sig:2004 Subsig:

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